Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Computer Vision
and Geometry Lab

Guided Monocular Depth Estimation

Master's Thesis

Sophie Selgrad
Department of Computer Science

Advisors: Dr, Christoph Vogel and Dr, Silvano Galliani
Supervisor: Prof. Dr. Marc Pollefeys
Computer Vision and Geometry Group
Department of Computer Science D-INFK

Abstract

This thesis studies guided monocular depth estimation with sparse depth maps as conditions in a latent diffusion framework. The goal is to predict a dense, affine-invariant depth map from a single RGB image while respecting a sparse set of depth measurements, a setting that arises in robotics, AR, and mapping when only partial range data are available.

Existing latent diffusion models for monocular depth estimation achieve strong zero-shot generalization but ignore sparse inputs at inference time. Depth completion methods can incorporate sparse data, yet state-of-the-art approaches commonly rely on multi-step sampling and test-time optimization, which increases latency and complicates deployment. The technical gap is an inference-efficient model that leverages sparse depth guidance without per-image optimization.

We attach a ControlNet to an E2E-FT depth backbone and evaluate two designs. A naïve variant supplies the sparse depth image to a ControlNet while keeping the backbone frozen. An inpainting variant conditions in the VAE latent space by concatenating the RGB latent, an encoded sparse-depth latent, and a binary validity mask, and trains with a composite objective that combines the standard scale-and-shift invariant loss with a sparse-consistency term at measured pixels. Both models use a single deterministic denoising step.

Experiments on NYUv2, ScanNet, DIODE, and KITTI show that explicit sparse-depth conditioning improves over monocular diffusion baselines under the same zero-shot protocol. The inpainting variant yields consistently lower error and higher accuracy on indoor datasets while maintaining single-pass inference. Compared with optimization-based depth completion, accuracy is lower but inference is substantially faster due to the absence of test-time optimization.

These results indicate that pretrained diffusion backbones can be upgraded into efficient guided predictors by injecting sparse-depth signals directly into the denoiser and aligning conditioning with the latent prediction domain. The approach offers a practical speed–accuracy trade-off for applications that require stable latency, and it provides a foundation for future work on modest multi-step variants, broader training data, and mechanisms for recovering absolute scale.

Acknowledgements

I would like to thank my co-supervisors, Dr. Christoph Vogel and Dr. Silvano Galliani, for their consistent support, constructive ideas, and helpful suggestions throughout the course of this thesis. Their guidance and technical insight were essential to shaping the direction of this work and solving many of the practical challenges along the way. I also thank Prof. Dr. Marc Pollefeys for supervising this thesis and for fostering a research environment that encourages rigorous and creative exploration.

I am grateful to the Computer Vision and Geometry Group (CVG) and to ETH Zürich for providing the computational resources and technical support that made this research possible.

Contents

1 Introduction **1**

2 Related Work **3**
2.1 Image generation. 3
2.2 Diffusion-based Conditional Image Synthesis. 3
2.3 Monocular Depth Estimation. 5
2.4 Depth Completion 6

3 Background **7**
3.1 Denoising Diffusion Probabilistic Models 7
3.1.1 Forward Diffusion Process 7
3.1.2 Reverse Denoising Process 7
3.1.3 Training Objective 8
3.2 Latent Diffusion Models 8
3.3 Latent Diffusion for Monocular Depth Estimation 9
3.3.1 Marigold 9
3.3.2 E2E-FT 10
3.4 ControlNet 10
3.4.1 Structure of a ControlNet 11
3.4.2 ControlNet for Text-to-Image Diffusion 11
3.4.3 Training 12

4 Method **15**
4.1 Problem Statement 15
4.2 Training Naïve ControlNet Model for Guided MDE 16
4.2.1 Data preprocessing 16
4.2.2 Fine-tuning procedure 17
4.3 Inference with Naïve ControlNet Model 19
4.3.1 Limitations of the Naïve ControlNet Model 21
4.4 Training the ControlNet E2E FT Inpainting Model for Guided Monocular Depth Estimation 22
4.5 Inference with the ControlNet Inpainting Model 24

5 Experiments **27**
5.1 Evaluation datasets 27
5.1.1 Experimental Setup 27
5.2 Results 28

6 Discussion **31**

7 Conclusion **33**

A Appendix **35**
A.1 Full Experiment Results 35
A.1.1 Full Results on ScanNet 35
A.1.2 Full Results on NYUv2 35
A.1.3 Full Results on KITTI 35
A.1.4 Full Results on DIODE 36
A.2 Naive Experiment Results 36
A.2.1 DIODE (Scale 1.0) 37
A.2.2 KITTI (Scale 1.0) 37
A.2.3 NYUv2 (Scale 1.0) 37
A.2.4 ScanNet (Scale 1.0) 38

List of Figures

3.1 Comparison between a standard neural network block and its ControlNet-augmented counterpart. (a) Before: a single network block transforms input features $\mathbf{x}$ into output $\mathbf{y}$. (b) After: a locked copy of the original block is paired with a trainable branch connected via zero-convolution layers, which inject conditioning $\mathbf{c}$ without disturbing pretrained weights. . 11

3.2 Integration of ControlNet into the Stable Diffusion U-Net. (**a**) Stable Diffusion consists of an encoder, middle block, and decoder with skip connections. (**b**) ControlNet attaches trainable copies (blue) to the encoder and middle blocks, connected via zero-convolution layers (white). Conditional features $\mathbf{c}_f$ are injected through the ControlNet while the original backbone (gray) remains frozen. 13

List of Tables

5.1 **Quantitative comparison** of Marigold and other state-of-the-art affine-invariant depth estimators on several zero-shot benchmarks. All metrics† are presented in percentage terms; **bold** numbers denote the best performance and underlined numbers denote the second best amongst non-Marigold-DC frameworks. *Ours (Naive Implementation)* refers to the baseline described in §4.2, while *Ours (ControlNet E2E FT)* corresponds to the improved model introduced in §4.4. Our method achieves consistently strong results across indoor and outdoor benchmarks without training on real depth data. 29

A.1 Full quantitative results on the **ScanNet** validation set. The table reports ten standard depth estimation metrics, including absolute and squared relative differences, various RMSE variants, $\log_{10}$ error, and accuracy under thresholds δ_1–δ_3. 35

A.2 Full quantitative results on the **NYUv2** validation set. Reported metrics include absolute and squared relative errors, RMSE variants, $\log_{10}$ error, and accuracy thresholds δ_1–δ_3. . . . 36

A.3 Full quantitative results on the **KITTI** Eigen test split. Results include ten standard metrics for depth estimation, enabling a detailed comparison across methods. 36

A.4 Full quantitative results on the **DIODE** validation set. The table reports all standard evaluation metrics for depth estimation. 37

A.5 Naive model results on the **DIODE** validation set for different sparse input configurations. All experiments were conducted at Scale 1.0. 37

A.6 Naive model results on the **KITTI** Eigen test split for different sparse input configurations. All experiments were conducted at Scale 1.0. 38

A.7 Naive model results on the **NYUv2** validation set for different sparse input configurations. All experiments were conducted at Scale 1.0. 38

A.8 Naive model results on the **ScanNet** validation set for different sparse input configurations. All experiments were conducted at Scale 1.0. 39

Chapter 1

Introduction

Monocular depth estimation (MDE) predicts a dense depth map from a single RGB image [43]. Depth completion (DC) instead densifies a sparse depth map, typically in metric units, using the accompanying RGB image as guidance [92]. Both are central to robotics and autonomous driving (navigation, obstacle avoidance, 3D mapping), AR/VR (scene understanding and occlusion), and visual effects and photography (relighting, refocusing) [20]. In many practical systems only sparse range measurements are available at test time, for instance from low-cost LiDAR or structure-from-motion (SfM) pipelines that yield a small set of reliable 3D points. This creates a need for *guided* depth prediction: using the strong image prior of MDE while enforcing consistency with sparse depth observations.

Recent latent diffusion models (LDMs) adapted for MDE have markedly improved zero-shot generalization across indoor and outdoor images. Marigold [43, 44] repurposes Stable Diffusion [75] for affine-invariant MDE by operating entirely in the latent space with a frozen VAE and a retrained U-Net, producing detailed depth with robust cross-dataset transfer [43]. E2E-FT revisits this setup and shows that correct trailing timestep scheduling allows for single-step inference, and that end-to-end fine-tuning at the final timestep yields deterministic predictions that match or exceed multi-step sampling while being orders of magnitude faster [20]. In parallel, Murre demonstrates that conditioning diffusion models on sparse SfM 3D points can anchor monocular predictions effectively, further demonstrating the value of integrating sparse geometric evidence during generation [27]. These advances suggest a promising path: retain the generalization of LDM-based MDE while injecting sparse depth constraints directly into the generative process.

Two broad strategies exist for using constraints with diffusion backbones. One may remain agnostic during generation and reconcile constraints post hoc, for example by affine alignment or test-time optimization on decoded predictions [105]. Marigold-DC follows this route: it keeps the pretrained MDE prior fixed and, at each denoising step, previews the clean latent, decodes to pixel space, fits scale and shift, and back-propagates a sparse-point loss to steer the latent and affine parameters [92]. This achieves strong zero-shot DC but requires many denoising steps, careful optimization hyperparameters, and often ensembling, which increases runtime. Alternatively, one can impose constraints inside the denoising network so that a single forward pass already accounts for the guidance. ControlNet provides such a mechanism for Stable Diffusion by attaching a trainable conditional branch that injects spatial features into the frozen backbone through zero-initialized adapters, enabling explicit control from images such as edges, poses, or depth [106].

This thesis explores guided monocular depth estimation with sparse depth maps as conditions by augmenting an E2E-FT depth backbone with ControlNet. We consider two designs. A naïve variant supplies the sparse depth image to a ControlNet while keeping the Stable Diffusion VAE and U-Net frozen, aiming to preserve the deterministic single-step behavior of E2E-FT. An inpainting variant makes the conditioning explicit in the latent domain by concatenating the RGB latent, an encoded sparse-depth latent, and a binary validity mask, and by training with a composite loss that combines the affine-invariant objective with

a sparse-consistency term on measured pixels. Both variants target deterministic, single-pass inference that take into consideration sparse inputs while preserving the strong image prior of LDMs. Our contributions are:

- We formulate guided, affine-invariant depth prediction in a latent-diffusion framework by attaching a ControlNet to an E2E-FT depth model, enabling explicit conditioning on sparse depth with single-step inference.
- We introduce an inpainting-style conditioning branch that operates in the VAE latent space and uses a binary validity mask, together with a training objective that couples the standard scale-and-shift invariant loss with a sparse-consistency penalty on observed pixels.
- We analyze why a naïve ControlNet attachment provides limited gains and show how latent-space alignment and masked conditioning remedy these issues.
- We evaluate on the standard zero-shot MDE splits (NYUv2, ScanNet, DIODE, KITTI) to compare against monocular LDM baselines and on the same data against Marigold-DC to study the accuracy–efficiency trade-off. Our method improves over monocular LDM estimators when sparse guidance is available and runs in a single forward pass, offering a favorable speed profile compared to multi-step test-time optimization.

The remainder of this thesis is structured as follows. Chapter 2 reviews prior work on diffusion-based image synthesis, monocular depth estimation, and depth completion, with emphasis on recent latent diffusion approaches. Chapter 3 reviews important background knowledge regarding LDMs, Marigold, E2E-FT, and ControlNet. Chapter 4 introduces our method in detail, covering the problem formulation, network architecture, training procedure, and inference pipeline. Chapter 5 presents the experimental setup, evaluation datasets, and quantitative comparisons with baseline methods. Chapter 6 interprets the findings, discusses their implications and limitations, and outlines potential directions for future work.

Chapter 2

Related Work

The goal of this work is guided depth estimation or depth completion using diffusion models conditioned on sparse depth inputs. We first review progress in image generation that led to the current latent diffusion architectures forming the basis of our method.

2.1 Image generation.

Early generative approaches such as VAEs [45] and autoregressive models [90] model data likelihoods explicitly but are limited by blurry reconstructions or slow, sequential sampling. Adversarial training in GANs [38, 39] enabled high-fidelity synthesis but remains unstable and prone to mode collapse.

Diffusion probabilistic models [83, 30] replaced adversarial training with a denoising process that learns to invert progressive noise corruption. Subsequent work improved efficiency and image quality through parameterized variance and cosine noise schedules [64], and by introducing non-Markovian deterministic sampling in DDIMs [84]. Diffusion models were later shown to outperform GANs on ImageNet using classifier guidance [16], while classifier-free guidance [32] simplified conditioning and improved controllability.

Latent Diffusion Models (LDMs) made diffusion approaches computationally practical by performing denoising in a compressed latent space obtained from a perceptual autoencoder. Operating in this domain preserves semantic structure while reducing cost, and cross-attention layers enable conditioning on text, segmentation maps, or other modalities. Stable Diffusion [75], trained on LAION-5B [81], extended this formulation to large-scale text-to-image generation using classifier-free guidance. In this work, text prompts are disabled (an empty prompt is used), and conditioning is provided entirely through image-based inputs.

The development of latent diffusion models established a stable, scalable framework for high-fidelity synthesis with explicit conditioning. Their ability to encode strong photometric and geometric priors makes them a natural foundation for tasks such as monocular depth estimation and depth completion.

2.2 Diffusion-based Conditional Image Synthesis.

Diffusion-based conditional image synthesis (DCIS) refers to the use of diffusion models for generating images that satisfy user-provided conditions by integrating control signals into the diffusion process, either within the denoising network or during sampling [105]. DCIS frameworks can be grouped by how the conditioning is introduced: into the denoising network through training [75, 78, 65, 69, 1, 24], re-purposing [63, 106, 93, 48, 18, 46], or specialization [42, 19, 96, 60, 72, 107, 3, 76, 25], and into the sampling process through inversion [61, 86, 62, 95, 34], attention manipulation [1, 29, 88], noise blending [53, 32], process

revision [57, 41, 40], guidance [16, 13, 67], or conditional correction [13, 85, 56, 11, 14]. Early training-based approaches condition models trained from scratch for image restoration tasks such as SR3 [79], CDM [31], and Palette [77]. These methods concatenate degraded inputs directly with noise or image tensors during training but do not leverage large text-to-image (T2I) backbones, making them less relevant for settings where pre-trained latent diffusion models are adapted for new visual control signals.

Re-purposing approaches instead adapt an existing T2I backbone with lightweight conditional encoders. T2I-Adapter [63] adds a convolutional encoder and injects multi-scale features into the U-Net to handle visual signal to image tasks. IP-Adapter [100] and GLIGEN [49] modify or add attention modules for layout control and image composition. PITI [93] maps images to the text embedding space to perform image-to-image translation. These methods sit close to our approach architecturally, as they attach an additional encoder to a frozen or partially frozen latent diffusion model and inject condition features into its intermediate layers.

Specialization methods fine-tune larger parts of the diffusion backbone to handle complex or multi-modal conditions. InstructPix2Pix [4] fine-tunes a text-to-image model for instruction-based editing by using paired before/after image data. Customization-oriented methods such as Imagic [42], Textual Inversion [19], and DreamBooth [76] focus on capturing user-specific concepts or objects, updating a small subset of model parameters or embeddings while preserving the model's generalization ability. These approaches enable personalized synthesis but at the cost of additional training and reduced modularity.

Alongside these training-time frameworks, several sampling-time control techniques have been developed. Inversion-based methods such as SDEdit [61] and DDIM [84] or Null-text inversion [62] reconstruct the latent corresponding to a given image and modify it to enable image editing. Attention manipulation methods like Prompt-to-Prompt [29], PnP [88], and MasaCtrl [6] alter or localize attention maps to control layout and semantics. Noise blending and process-revising approaches, including Composable Diffusion Models [53], classifier-free guidance [32], IR-SDE [57], and DDRM [40], combine or alter diffusion processes to balance multiple conditions or solve inverse problems. Guidance and correction methods such as Classifier Guidance [16], DPS [12], Pix2Pix-Zero [67],

The most influential development in this line is ControlNet [106], which augments Stable Diffusion by cloning its encoder and middle block and connecting the cloned branch to the frozen backbone through zero-initialized 1×1 convolutions. This design allows ControlNet to learn spatial conditioning while preserving the base model's pretrained knowledge. Condition features are injected at matched resolutions through the U-Net skip connections, allowing for fine-grained control over structure and geometry. ControlNet supports various types of conditions, including Canny and HED edges, M-LSD lines, monocular depth, surface normals, semantic segmentation (ADE20K [108]), human pose (OpenPose [7]), and sketches. Multiple conditions can be applied concurrently by summing their corresponding feature maps. It also introduces classifier-free guidance resolution weighting for stable prompt-free control and drops text inputs 50% of the time during training to reinforce purely visual conditioning. These properties make ControlNet data-efficient and robust, achieving competitive depth-to-image performance with around 200k images on a single 3090Ti GPU.

In this work, we adopt ControlNet as the conditioning framework for latent diffusion models. Its design aligns naturally with our objective of using sparse depth maps as control signals for depth completion, leveraging pre-trained diffusion priors while enabling spatially grounded conditioning without retraining a full generative model.

2.3 Monocular Depth Estimation.

Monocular depth estimation (MDE) aims to infer scene geometry from a single RGB image. Predicted depths are typically categorized by the type of scale information they preserve. Relative depth predicts only depth ordering (which pixel is closer), as in DIW-style evaluations with sparse pairs; no global scale or shift is defined [99]. Affine-invariant depth is defined up to a global scale and shift, i.e., $d' = a\,d + b$ with $a > 0$. Training and evaluation use scale- and shift-invariant objectives, and the representation is independent of focal length while preserving meaningful depth ratios across the scene [71, 43, 44, 99]. Metric depth predicts absolute distance in physical units under the test camera intrinsics; it enables metrology and SLAM without external scale recovery but is sensitive to focal length and camera model, motivating explicit handling of intrinsics as in Metric3D's canonical camera transformation [101].

In this work we estimate *affine-invariant* depth, following prior practice for in-the-wild generalization and for compatibility with latent-diffusion training and evaluation [71, 43, 44, 99].

Early MDE methods trained convolutional networks on mixed indoor–outdoor RGB–D datasets to improve cross-domain generalization. MiDaS and DPT demonstrated that large-scale, affine-invariant training enables zero-shot transfer but yields depth maps defined only up to an affine transformation, limiting metric consistency across cameras [71, 70, 17, 102]. To recover metric scale, Metric3D introduced canonical camera-space normalization and mixed-camera supervision, achieving consistent metric predictions across varied intrinsics and improved zero-shot performance on in-the-wild images [101].

Recent foundation-style models pursue large-scale pretraining and unlabeled data integration. Depth Anything (DA) uses a DINOv2 encoder and a teacher–student pipeline trained on 1.5M labeled and 62M pseudo-labeled images, combining affine-invariant and semantic-alignment losses to enhance structural detail and generalization [98]. Depth Anything v2 extends this framework using synthetic supervision for all labeled data and scaling to billions of parameters, yielding finer detail, better handling of transparent and reflective surfaces, and faster inference across model sizes [99]. These models remain strong discriminative baselines for zero-shot MDE.

Parallel work has explored generative diffusion frameworks for dense prediction. Lotus applies diffusion-based learning to dense estimation tasks and studies the effect of prediction parameterization and denoising steps, showing that generative priors can improve structure preservation in pixel-level outputs [28]. Although not limited to depth, this line of work motivates adapting latent diffusion backbones for geometric prediction.

Marigold fine-tunes Stable Diffusion v2 for MDE by disabling text conditioning and retraining the UNet to predict latent-space depth maps while keeping the VAE frozen. The model is trained on synthetic RGB–D data (Hypersim, Virtual KITTI2) using affine-invariant normalization and performs denoising in latent space. It achieves strong zero-shot results through probabilistic sampling and test-time ensembling, demonstrating that diffusion priors encode useful geometric information even without real data [43]. The follow-up study [44] introduces implementation refinements, including an updated DDIM schedule with trailing timesteps, improving stability for deterministic inference.

E2E-FT revisits Marigold's design and shows that the commonly used leading DDIM schedule misaligns timesteps and noise levels, producing inconsistent single-step predictions. By adopting the trailing schedule, the model can generate valid one-step outputs without sampling. E2E-FT fine-tunes the diffusion backbone end-to-end at $t{=}T$ with a deterministic regression loss, removing stochastic sampling while retaining geometric fidelity. The resulting model achieves comparable or better accuracy than Marigold's multi-step variants while being over 200× faster. The work also reports similar gains on related diffusion geometry estimators such as GeoWizard and influences subsequent Marigold releases, which incorporate the corrected scheduler [20].

Together, these developments bridge discriminative and generative paradigms. Metric3D and Depth Anything define the discriminative state of the art for large-scale MDE, while Marigold and E2E-FT establish diffusion-based alternatives that combine geometric reasoning with strong pre-trained visual priors. This progression motivates our approach of adapting latent diffusion models for depth completion using sparse-depth conditioning.

2.4 Depth Completion

Depth completion densifies sparse depth measurements into a full depth map using an accompanying RGB image as guidance. Unlike monocular depth estimation (MDE), which predicts depth from RGB alone, depth completion anchors predictions to measured points (often in metric units) and must handle sparsity, irregular sampling, and sensor noise.

Early approaches relied on sparse depth only, using classical interpolation or sparsity-invariant CNNs to avoid collapsing features at missing pixels [89]. With RGB guidance, encoder–decoder fusion networks improved edge fidelity and extrapolation [59, 58, 52, 87, 68]. Subsequent work added multi-scale objectives [47], intermediate cues such as surface normals or coarse depth [68], and graph-based reasoning [97]. Spatial propagation networks further refined outputs: CSPN and CSPN++ introduced fixed and adaptive propagation kernels, NLSPN added non-local neighborhoods, DySPN learned dynamic affinities, and LRRU varied kernel scopes for long/short-range updates [10, 9, 66, 51, 94].

Transformers and geometry-aware designs broadened the model class. GuideFormer applied ViTs for image-guided completion; CompletionFormer combined convolutions and vision transformers; PointDC aggregated feature point clouds; geometry-aware models infused 3D structure or tri-perspective cues to improve layout and far-plane recovery [73, 103, 104, 8]. For stronger generalization, SpAgNet handled highly sparse/irregular inputs, VPP4DC revisited completion as fictitious stereo matching to reduce domain bias, OGNI-DC optimized a depth gradient field with iterative updates, and Prompt Depth Anything adapted a feed-forward depth foundation model to sparse prompting [2, 2, 109, 50].

Generative alternatives followed. Flow-matching and diffusion-based variants conditioned on RGB and densified sparse depth before or during processing (often via distance-function or latent-space interpolation), improving detail but introducing sensitivity to densification artifacts [26, 54, 23, 36]. These methods highlight a trade-off between using a strong prior and avoiding corruption around guidance points.

Marigold-DC reframes depth completion as image-conditional depth generation guided by sparse measurements [92]. It leverages a pretrained latent diffusion model for MDE (Marigold) as the visual prior and injects sparse depth at test time without fine-tuning the backbone. At each denoising step, the method previews the clean depth latent, decodes to pixel space, aligns to metric units via learnable scale/shift, and backpropagates a loss on valid sparse points to update the latent and affine parameters; a trailing DDIM schedule and test-time ensembling improve stability and robustness. This design exploits the generalization of the MDE prior while anchoring predictions to sparse measurements, yielding strong zero-shot performance across indoor/outdoor datasets and varying sparsity patterns [92].

In summary, classical fusion and propagation methods established reliable pipelines under fixed conditions [59, 10, 66, 103], generalization-oriented designs improved cross-domain transfer [2, 109], and generative variants explored stronger priors [26, 54]. Marigold-DC combines these threads by using a diffusion-based monocular prior as the scene model and incorporating sparse depth as test-time guidance, which aligns with our broader strategy of conditioning latent diffusion on sparse depth for robust completion [92].

Chapter 3

Background

3.1 Denoising Diffusion Probabilistic Models

Diffusion models are a class of generative models that learn to approximate a complex data distribution by gradually reversing a fixed noising process.

3.1.1 Forward Diffusion Process

Denoising Diffusion Probabilistic Models (DDPMs) are trained to reverse a stochastic forward process [30]. This process, the approximate posterior $q(\mathbf{x}_{1:T} \mid \mathbf{x}_0)$, forms a Markov chain that transforms a data sample $\mathbf{x}_0 \sim q(\mathbf{x}_0)$ into increasingly noisy versions $\mathbf{x}_1, \ldots, \mathbf{x}_T$ by sequentially adding Gaussian noise according to a variance schedule $\beta_1, \ldots, \beta_T$:

$$q(\mathbf{x}_{1:T} \mid \mathbf{x}_0) := \prod_{t=1}^{T} q(\mathbf{x}_t \mid \mathbf{x}_{t-1}), \qquad q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}\Big(\mathbf{x}_t; \sqrt{1-\beta_t}\,\mathbf{x}_{t-1},\ \beta_t \mathbf{I}\Big)\,. \tag{3.1}$$

Using the cumulative product of noise coefficients $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$ and defining $\alpha_t := 1 - \beta_t$, this can be expressed in closed form as

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}\big(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0,\ (1-\bar{\alpha}_t)\mathbf{I}\big)\,. \tag{3.2}$$

With a sufficiently large number of diffusion steps T, the terminal distribution $q(\mathbf{x}_T)$ approaches an isotropic Gaussian. A small noise variance β_t allows the reverse process to be approximated as Gaussian as well.

3.1.2 Reverse Denoising Process

The objective of training is to learn the reverse denoising process $p_\theta(\mathbf{x}_{0:T})$, a Markov chain that reconstructs data $\mathbf{x}_0$ from noisy inputs by iteratively predicting and removing the added noise. Starting from $p(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I})$, the reverse transitions are parameterized as

$$p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t))\,, \tag{3.3}$$

where the mean μ_θ and variance Σ_θ are represented by neural networks. Other works have modified this formulation to significantly reduce the number of diffusion or inference steps T needed to generate high-quality samples [37, 64, 84].

3.1.3 Training Objective

Training minimizes the variational lower bound (VLB) on the negative log-likelihood of the data. The overall objective can be expressed as

$$\mathbb{E}_q[-\log p_\theta(\mathbf{x}_0)] \leq \mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T} \mid \mathbf{x}_0)}\right]. \tag{3.4}$$

This bound decomposes into a sum of Kullback–Leibler (KL) divergence terms between the learned reverse process $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ and the forward process posterior $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)$, plus a reconstruction term on the data. Since all involved distributions are Gaussian, each KL divergence can be computed in closed form.

Following Ho et al. [30], the variance $\Sigma_\theta(\mathbf{x}_t, t)$ is fixed, and the model is trained to predict the noise term $\boldsymbol{\epsilon}$ used to corrupt the data in the forward process:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t,\,\mathbf{x}_0,\,\boldsymbol{\epsilon}\sim\mathcal{N}(0,I)}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right\|_2^2\right], \qquad \mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}. \tag{3.5}$$

This simple noise-prediction loss yields stable optimization and high-quality samples. During inference, samples are generated by starting from $\mathbf{x}_T \sim \mathcal{N}(0, I)$ and iteratively denoising through the learned reverse transitions. A deterministic alternative, the Denoising Diffusion Implicit Model (DDIM) [84], reformulates the reverse process as a non-Markovian update rule that preserves the same training objective but allows sampling in far fewer steps.

3.2 Latent Diffusion Models

Diffusion in pixel space is computationally demanding. Latent Diffusion Models (LDMs) address this limitation by performing the diffusion process in a compressed latent space obtained from a pretrained variational autoencoder (VAE). Given an image $\mathbf{x}$, the encoder $\mathcal{E}$ maps it to a latent representation $\mathbf{z}_0 = \mathcal{E}(\mathbf{x})$, and the decoder $\mathcal{D}$ reconstructs it as $\mathcal{D}(\mathbf{z}_0) \approx \mathbf{x}$. The diffusion model then learns the reverse process in this latent domain:

$$q(\mathbf{z}_t \mid \mathbf{z}_0) = \mathcal{N}\big(\mathbf{z}_t; \sqrt{\bar{\alpha}_t}\,\mathbf{z}_0,\, (1-\bar{\alpha}_t)\mathbf{I}\big), \qquad p_\theta(\mathbf{z}_{t-1} \mid \mathbf{z}_t) = \mathcal{N}(\mathbf{z}_{t-1}; \mu_\theta(\mathbf{z}_t, t), \Sigma_\theta(\mathbf{z}_t, t)). \tag{3.6}$$

By operating in a lower-dimensional representation, the model captures high-level semantics while reducing memory and computational cost. The latent-space formulation also improves stability by abstracting away pixel-level noise and focusing on perceptually meaningful structure.

In diffusion-based image generation, the optimal prediction at the final timestep T corresponds to the mean of the data distribution. For text-conditional models this mean tends to blur multiple modes, but for image-conditioned tasks such as monocular depth estimation, the conditional distribution is usually unimodal—each image corresponds to a single ground-truth depth map. Consequently, diffusion-based latent models are well suited for deterministic prediction tasks.

Stable Diffusion [75] is a widely used example of an LDM. It employs a convolutional U-Net that performs denoising in the latent space and conditions on text features produced by a CLIP text encoder. This latent formulation enables efficient generation of high-resolution images and forms the basis for many extensions that incorporate additional conditioning signals such as edges, segmentation masks, or depth maps. In this work, we incorporate sparse depth maps as an additional conditioning signal.

With the success of Marigold [43], models that adapt pretrained text-to-image latent diffusion architectures for tasks such as monocular depth estimation and depth completion have gained traction. These

approaches leverage the strong visual priors of large pretrained diffusion models while remaining computationally practical compared to transformer-based depth models such as Depth Anything [98] and Depth Anything V2 [99]. Following this direction, Garcia et al. [20] showed that end-to-end fine-tuning of a pretrained image-conditional LDM with task-specific losses provides an effective framework for monocular depth and normal estimation. The following sections review Marigold and E2E-FT, two LDM-based MDE frameworks, in more detail.

3.3 Latent Diffusion for Monocular Depth Estimation

3.3.1 Marigold

In the context of MDE, Marigold [43, 44] repurposes Stable Diffusion v2 as an image-conditional LDM. Instead of conditioning on text, the diffusion process is conditioned on an RGB image $\mathbf{i}_{\text{RGB}} \in \mathbb{R}^{H\times W\times 3}$ to generate its corresponding depth map $\mathbf{d}^* \in \mathbb{R}^{H\times W}$. Thus, Marigold is trained to model the conditional distribution $D(\mathbf{d}^* \mid \mathbf{i}_{\text{RGB}})$.

Monocular depth estimation can be formulated as a diffusion-based image-conditional generation problem in which the conditional distribution is close to unimodal, since each image corresponds to a unique depth map. Consequently, the model's prediction at the final timestep T is expected to align closely with the true depth, making diffusion-based latent models well suited for deterministic prediction.

During training, the ground-truth depth map $\mathbf{d}^*$ is normalized to roughly $[-1, 1]$. Because the pretrained VAE encoder expects three input channels, the single-channel depth map is duplicated across channels to form a three-channel input, which is then encoded to produce the depth latent $\mathbf{z}^* = \mathcal{E}(\mathbf{d}^*)$. A random timestep t is sampled, and Gaussian noise is added to obtain the noised depth latent:

$$\mathbf{z}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{z}^* + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \tag{3.7}$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The RGB image is also encoded as $\mathbf{x} = \mathcal{E}(\mathbf{i}_{\text{RGB}})$, which remains noise-free to serve as the conditioning signal.

To allow the RGB latent to condition the denoiser, Marigold concatenates $\mathbf{x}$ and $\mathbf{z}_t$ along the channel dimension, doubling the input channels of the first convolution. To maintain the pretrained network's scale, the weight tensor of the first convolutional layer is duplicated and scaled by $1/2$. Marigold also employs v-parameterization [80], predicting

$$\hat{\mathbf{v}}_t = \mathbf{v}_\theta([\mathbf{z}_t, \mathbf{x}], t), \tag{3.8}$$

where $\mathbf{v}_\theta$ is the denoising network. The training objective minimizes the mean-squared error between $\hat{\mathbf{v}}_t$ and the ground-truth target

$$\mathbf{v}_t^* = \sqrt{\bar{\alpha}_t}\,\boldsymbol{\epsilon} - \sqrt{1-\bar{\alpha}_t}\,\mathbf{z}^*, \tag{3.9}$$

yielding

$$\mathcal{L} = \mathbb{E}_{\mathbf{d}^*,\,\boldsymbol{\epsilon}\sim\mathcal{N}(\mathbf{0},\mathbf{I}),\,t\sim\mathcal{U}\{1,\ldots,T\}}\left[\left\|\hat{\mathbf{v}}_t - \mathbf{v}_t^*\right\|_2^2\right]. \tag{3.10}$$

During inference, a Gaussian noise latent $\mathbf{z}_T$ is concatenated with the RGB latent $\mathbf{x}$ and repeatedly denoised using $\mathbf{v}_\theta$ (e.g., for 50 DDIM steps [84]). The final latent is decoded as $\hat{\mathbf{d}} = \mathcal{D}(\mathbf{z}_0)$, and its three channels are averaged to obtain the predicted depth map.

Because initialization with random noise introduces stochasticity, Marigold averages multiple affine-aligned predictions at test time for stability. In their ensemble scheme [43], N samples are generated, aligned by estimated scale $\hat{s}_i$ and shift $\hat{t}_i$, and combined via a pixel-wise median. This yields smoother, more consistent results. Additional improvements include multi-resolution noise and isotropic Gaussian

sampling during training. Overall, Marigold serves as an image-conditional latent diffusion network built on a pretrained Stable Diffusion backbone with a frozen VAE encoder–decoder pair $(\mathcal{E}, \mathcal{D})$ and a retrained U-Net.

3.3.2 E2E-FT

Garcia et al. [20] build upon Marigold by addressing two key limitations: the lack of valid single-step inference and the use of random noise initialization. They identify and correct a scheduling inconsistency that prevented deterministic inference and replace Gaussian initialization with a zero tensor for stability. Additionally, they introduce a task-specific objective to improve depth accuracy.

Fixing Single-Step Inference

In Marigold, the denoising process typically requires 50 steps. Garcia et al. observed that the predicted depth changes little across these steps, suggesting that one-step inference should suffice. However, the single-step output resembled noise due to a scheduler misconfiguration. Marigold used the default `leading` timestep spacing [35], creating a mismatch between the timestep encoding and actual noise level. The model therefore received an encoding corresponding to an almost clean latent while the input was still pure noise.

To fix this, E2E-FT employs the `trailing` timestep spacing for both training and inference, aligning timestep encoding with the true noise level. This correction enables accurate single-step predictions. Following this update, the Marigold authors released improved versions (e.g., Marigold-HR) adopting this scheduler alignment.

End-to-End Fine-Tuning Procedure

After correcting the scheduler, Garcia et al. fine-tune the model end-to-end. Naïve multi-step finetuning would be prohibitively expensive, so they fix $t = T$ during training, explicitly optimizing for single-step prediction. The Gaussian-initialized latent is replaced by a zero tensor, since $\mathbf{0}$ is the mean of the standard normal distribution. The model thus learns a deterministic mapping from $\mathbf{x}$ to $\mathbf{z}^*$:

$$\mathbf{v}_T^* = \sqrt{\bar{\alpha}_T}\,\boldsymbol{\epsilon} - \sqrt{1-\bar{\alpha}_T}\,\mathbf{z}^*, \tag{3.11}$$

and

$$\hat{\mathbf{z}}_0 = \sqrt{\bar{\alpha}_t}\,\mathbf{z}_t - \sqrt{1-\bar{\alpha}_t}\,\hat{\mathbf{v}}_\theta([\mathbf{z}_t, \mathbf{x}], t)\,. \tag{3.12}$$

Unlike Marigold, which minimizes MSE on latent predictions, E2E-FT optimizes a task-specific loss on the decoded depth maps, producing deterministic inference aligned with the downstream task. For depth estimation, the affine-invariant loss is used:

$$\mathcal{L}_{\mathrm{D}} = \frac{1}{HW}\sum_{i,j}\left|d_{i,j}^* - \hat{d}_{i,j}\right|, \tag{3.13}$$

where $\hat{d}_{i,j} = sd_{i,j}+t$ denotes the least-squares aligned prediction. We use the publicly released `GonzaloMG/stable-dif` checkpoint [22] as the pretrained base for our custom ControlNet inpainting model.

3.4 ControlNet

While E2E-FT enables efficient single-step inference, it remains monocular and cannot directly exploit sparse-depth guidance. To enable explicit conditioning, we adopt the ControlNet framework [106], described below.

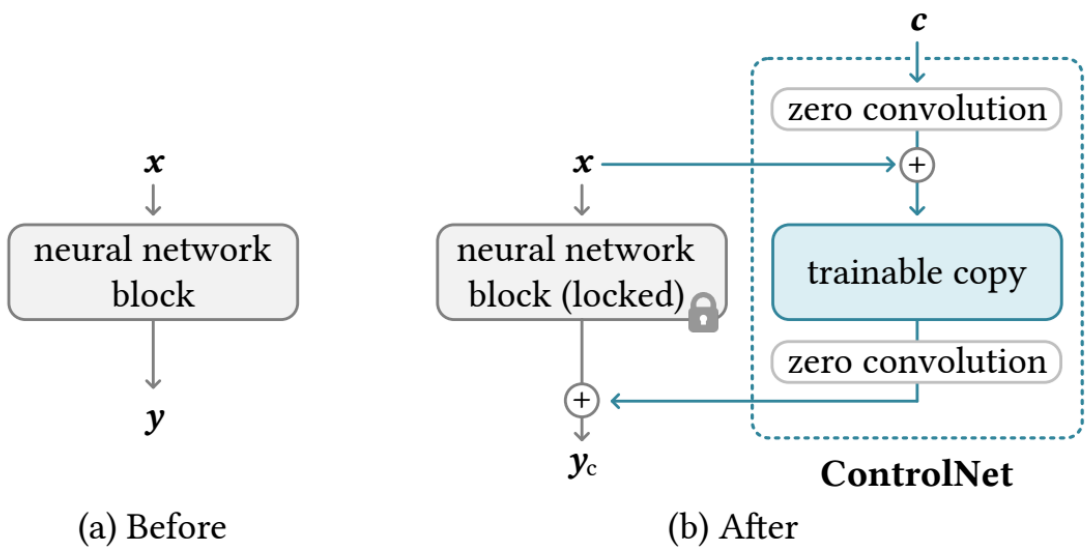

Figure 3.1: Comparison between a standard neural network block and its ControlNet-augmented counterpart. (a) Before: a single network block transforms input features $\mathbf{x}$ into output $\mathbf{y}$. (b) After: a locked copy of the original block is paired with a trainable branch connected via zero-convolution layers, which inject conditioning $\mathbf{c}$ without disturbing pretrained weights.

3.4.1 Structure of a ControlNet

ControlNet augments a pretrained diffusion model by introducing a secondary, trainable branch that injects spatially localized conditioning information into intermediate network blocks (Figure 3.1). A *network block* refers to a sequence of layers, such as a residual block, a convolution–BN–ReLU unit, or a transformer attention module. Let $F(\cdot;\Theta)$ denote a pretrained block with parameters Θ, transforming input $\mathbf{x}$ into output $\mathbf{y}$:

$$\mathbf{y} = F(\mathbf{x};\Theta). \tag{3.14}$$

A trainable copy with parameters Θ_c accepts an external conditioning vector $\mathbf{c}$. The two branches are connected via *zero-convolution* layers, $Z(\cdot;\cdot)$, which are 1×1 convolutions initialized to zero. With Θ_{z1} and Θ_{z2} denoting their parameters, a ControlNet block computes

$$\mathbf{y}_c = F(\mathbf{x};\Theta) + Z(F(\mathbf{x} + Z(\mathbf{c};\Theta_{z1});\,\Theta_c)\,;\,\Theta_{z2})\,. \tag{3.15}$$

Initially, both zero-convs output zero, so $\mathbf{y}_c = \mathbf{y}$, ensuring no perturbation of pretrained activations. Over time, $Z(\cdot)$ learns non-zero weights that inject conditioning signals while keeping the backbone stable.

3.4.2 ControlNet for Text-to-Image Diffusion

Stable Diffusion [75] follows a U-Net architecture consisting of an encoder, middle block, and decoder with skip connections (Figure 3.2a). ControlNet attaches trainable copies to the encoder and middle blocks (Figure 3.2b), injecting conditioning features $\mathbf{c}_f = \mathcal{E}(\mathbf{c}_i)$ computed by a small convolutional encoder. This encoder consists of four 4×4 convolutions with stride 2, ReLU activations, and channel sizes 16, 32, 64, and 128. The resulting 64×64 feature map is fed into the ControlNet modules, influencing the diffusion process without altering the pretrained decoder.

This configuration is computationally efficient: since the locked backbone does not require gradients,

ControlNet training increases GPU memory usage by roughly 23% and iteration time by about 34% compared with finetuning Stable Diffusion alone.

3.4.3 Training

During training, the denoising model predicts the added noise $\boldsymbol{\epsilon}$ conditioned on the timestep t, a text embedding $\mathbf{c}_t$ (if any), and the task-specific feature map $\mathbf{c}_f$. The objective is the standard diffusion loss:

$$\mathcal{L} = \mathbb{E}_{\mathbf{z}_0,\, t,\, \mathbf{c}_t,\, \mathbf{c}_f,\, \boldsymbol{\epsilon}\sim\mathcal{N}(0,1)} \left[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t, \mathbf{c}_t, \mathbf{c}_f)\|_2^2 \right]. \tag{3.16}$$

When finetuning Stable Diffusion, 50% of text prompts are replaced with empty strings to encourage reliance on visual conditioning. In depth estimation, however, no text conditioning is used. Because the zero-conv layers start from zero, the pretrained backbone remains stable throughout optimization. ControlNet exhibits a "sudden convergence" behavior, where conditioning rapidly becomes effective after a few thousand steps. Additional mechanisms such as Classifier-Free Guidance Resolution Weighting (CFG-RW) and Multi-ControlNet composition were proposed in [106], but since this work focuses on single-condition image guidance without text prompts, these extensions are not used.

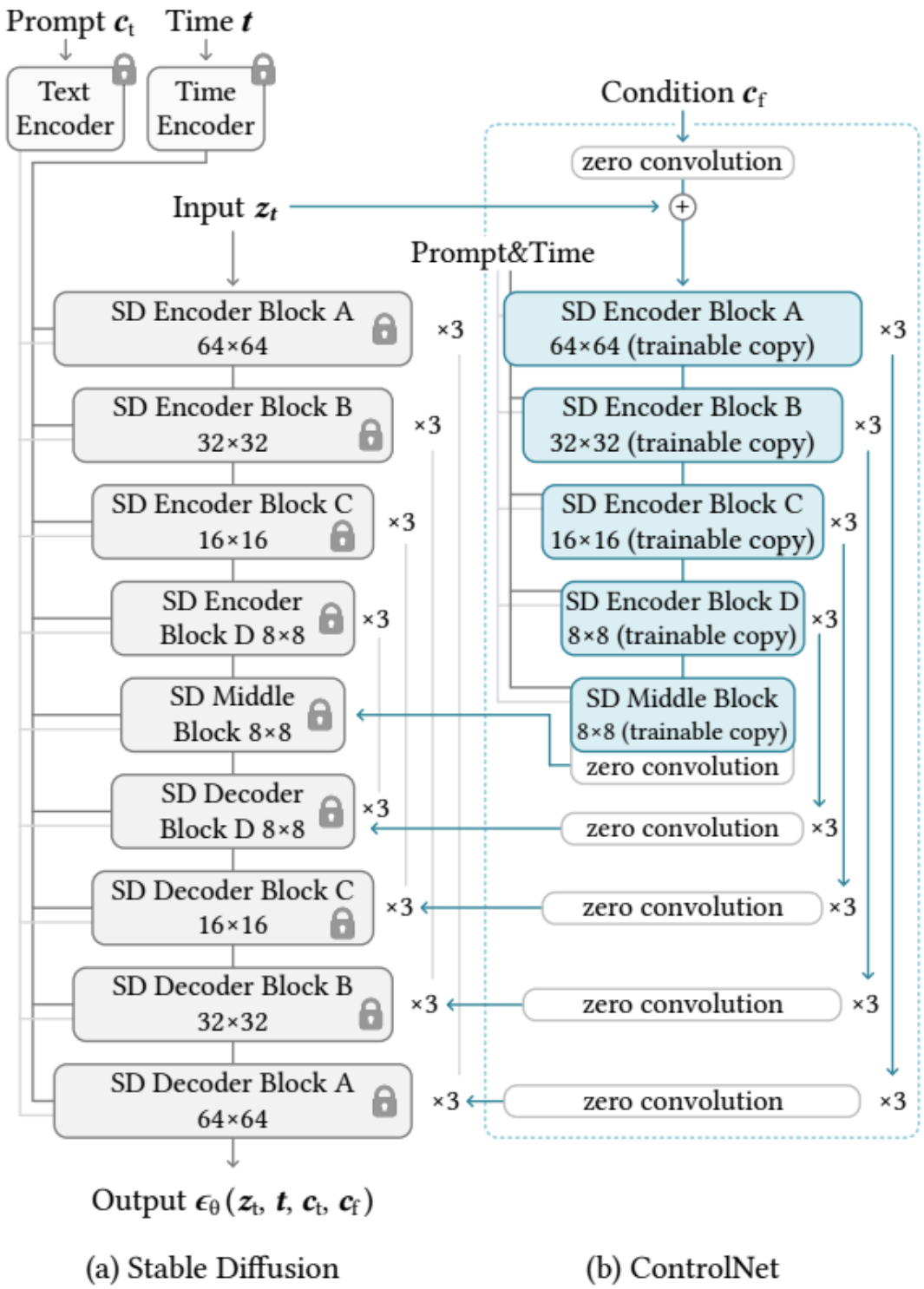

Figure 3.2: Integration of ControlNet into the Stable Diffusion U-Net. (**a**) Stable Diffusion consists of an encoder, middle block, and decoder with skip connections. (**b**) ControlNet attaches trainable copies (blue) to the encoder and middle blocks, connected via zero-convolution layers (white). Conditional features $\mathbf{c}_f$ are injected through the ControlNet while the original backbone (gray) remains frozen.

Chapter 4

Method

4.1 Problem Statement

The goal of this work is to perform *guided monocular depth estimation* using a diffusion-based framework. Given a single RGB image $\mathbf{i}_{\text{RGB}} \in \mathbb{R}^{H \times W \times 3}$ and a corresponding sparse depth map $\mathbf{d}_{\text{sparse}} \in \mathbb{R}^{H \times W}$, the objective is to predict an accurate dense depth map $\hat{\mathbf{d}} \in \mathbb{R}^{H \times W}$ that accurately captures the scene geometry and remains faithful to the sparse measurements. We assume that both camera intrinsics and extrinsics are unknown. Monocular also means that we assume that during prediction, we consider one pair consisting of a single RGB image and its corresponding sparse depth map pair at a time. This problem remains challenging due to the limited density of the sparse input and the inherent scale and shift ambiguity in monocular images.

Existing diffusion-based depth estimation models such as Marigold [43] and E2E-FT [20] operate in an image-conditional setting and do not incorporate sparse geometric guidance during inference. Marigold-DC [92] extends this paradigm by injecting sparse depth information into the denoising process through test-time optimization. While this approach demonstrates strong zero-shot generalization, it introduces several limitations: the optimization loop must backpropagate through the denoising process at every timestep, making inference computationally expensive; the method is highly sensitive to the initialization of the scale and shift parameters, which depend on representative sparse samples; and the stochasticity of diffusion sampling necessitates test-time ensembling to stabilize predictions, further increasing runtime. Because Marigold-DC optimizes the depth predictions only at inference time without retraining the diffusion backbone, the model does not learn to internally represent or generalize from sparse depth inputs. As a result, it cannot consistently exploit sparse depth maps across datasets or varying sparsity patterns.

To address these limitations, we build on the E2E-FT framework by training a ControlNet on top of the `GonzaloMG/stable-diffusion-e2e-ft-depth` checkpoint for monocular depth estimation. This enables us to condition the diffusion model explicitly on sparse depth maps rather than relying on test-time gradient updates. By incorporating the sparse depth signal as an auxiliary conditioning branch within the diffusion process, the proposed framework aims to achieve guided affine-invariant depth estimation with deterministic, single-pass inference. Our objective is to evaluate whether augmenting the E2E-FT depth model with sparse depth conditioning via a custom ControlNet improves accuracy and consistency compared to prior zero-shot optimization methods such as Marigold-DC, while substantially reducing inference time. We aim to create a framework for guided monocular depth estimation that is robust and can generalize well to out-of-training training samples.

We present two variants. The first, a naive ControlNet, feeds the raw sparse depth image (pixel space, unencoded) as the control image; the UNet input remains the RGB latent concatenated with a zero depth latent at $t{=}T$. The second, an inpainting variant, supplies the ControlNet with a concatenation of the RGB

latent, the encoded control image latent (from the sparse depth), and a binary mask, enabling the network to distinguish measured versus missing regions in latent space. Note that throughout this section, we refer to the base checkpoint `GonzaloMG/stable-diffusion-e2e-ft-depth` as `SD-E2E-FT-depth`.

4.2 Training Naïve ControlNet Model for Guided MDE

Our baseline applies a standard ControlNet on top of `SD-E2E-FT-depth` to enable conditioning on sparse depth. The RGB image $\mathbf{i}_{\text{RGB}} \in \mathbb{R}^{H\times W\times 3}$ is encoded by the frozen VAE to a latent $\mathbf{x}\in\mathbb{R}^{h\times w\times 4}$; a zero latent of the same shape serves as the predicted-depth channel at $t{=}T$, matching the single-step training of E2E-FT. The sparse depth map $\mathbf{d}_{\text{sparse}} \in \mathbb{R}^{H\times W}$ is provided to the ControlNet branch as the sole conditioning image, and the ControlNet outputs are injected into the UNet blocks of `SD-E2E-FT-depth`. Training follows the single-timestep regime (trailing spacing) with the same empty text embedding as the base model; inference uses a single denoising step and also uses zero initialization for the depth latent.

This variant is intentionally minimal: no architectural changes are made to the ControlNet embedding path beyond adapting channel counts on the UNet input to accommodate concatenated RGB and predicted-depth latents. We next describe the training setup for this baseline (datasets, preprocessing, and fine-tuning), and then the inference procedure. The inpainting variant follows.

4.2.1 Data preprocessing

Datasets. We train on two synthetic datasets with complementary scene coverage: Hypersim for indoor environments and Virtual KITTI 2 (VKITTI2) for driving scenes. The training split and preprocessing follow the same configuration as [20], using their provided download and extraction instructions. Synthetic depth offers dense, complete supervision without the missing/invalid pixels common in real sensors, which is an advantage when training in the Stable Diffusion VAE latent space, which assumes valid inputs everywhere [43]. Hypersim provides photorealistic indoor imagery with per-pixel geometry and rich scene annotations across hundreds of artist-authored scenes, making it well-suited for geometric learning and sim-to-real transfer [74]. VKITTI2 recreates KITTI [21] tracking sequences in a modern game engine, supplying pixel-perfect ground truth across weather and viewpoint variants (fog, rain, camera rotations), which helps cover outdoor structure and driving-specific layouts [5]. Following prior LDM-based depth work, we adopt the same Hypersim and VKITTI2 splits used by Marigold and E2E-FT (approximately $54{,}000$ indoor and $20{,}000$ outdoor frames, with far planes of 65 m and 80 m, respectively), ensuring a consistent experimental setup and comparable training conditions across models [20].

Sparse depth construction. Given a dense depth map $\mathbf{d}_{\text{dense}} \in \mathbb{R}^{H\times W}$, we form a sparse depth map by first drawing a per-image cardinality k from a discrete uniform distribution over a dataset-specific interval and then sampling k valid pixels uniformly at random. The selected pixels copy their original depth values; all others are set to zero, yielding $\mathbf{d}_{\text{sparse}}\in\mathbb{R}^{H\times W}$. For Hypersim we use $k \sim \text{Unif}\{\, k^{\text{H}}_{\min}, \ldots, k^{\text{H}}_{\max} \,\}$ with $k^{\text{H}}_{\min} = 40000$ and $k^{\text{H}}_{\max} = 320000$; for VKITTI2 we use $k \sim \text{Unif}\{\, k^{\text{V}}_{\min}, \ldots, k^{\text{V}}_{\max} \,\}$ with $k^{\text{V}}_{\min} = 20000$ and $k^{\text{V}}_{\max} = 320000$. Validity is defined by positive depth, optionally excluding the saturated far-plane code (e.g., 65535 for `uint16`) when present; no additional noise or perturbations are applied. This procedure is implemented in our dataset scripts and used for both Hypersim and VKITTI2 [1].
Why sample a range? Training across a range of k improves robustness to the unknown sparsity of test-time inputs and reduces overfitting to a fixed point density; it also better reflects practical scenarios where sensor coverage varies across scenes and frames. Empirically, exposing the ControlNet to multiple sparsity regimes

[1] See the code in `vkitti_sample_7-2-1-2.py` and `hypersim_sample_7-2-1-2.py`

stabilizes learning signals from the control branch (which would otherwise be tuned to a single density) and preserves performance when the number of available depth samples departs from the training median. The end-to-end pipeline and single-step training remain unchanged.

Data preprocessing. Our pipeline follows the affine-invariant normalization of Marigold [43] and E2E-FT [20], with extensions to handle sparse inputs. For each RGB–depth pair $(\mathbf{i}_{\text{RGB}},\ \mathbf{d}_{\text{dense}})$, validity is defined by the same near/far planes as E2E-FT: $[10^{-5},\ 65]$ m for Hypersim and $[10^{-5},\ 80]$ m for VKITTI2. Metric depth is first linearly mapped to $[0, 1]$ using these bounds. We then apply an ϵ-margin to *valid* pixels ($x \mapsto x(1-\epsilon)+\epsilon$, with $\epsilon = 0.02$), set *invalid* pixels to 0, and finally rescale to $[-1, 1]$. Under this transformation, valid depths occupy $[-0.96,\ 1]$, while invalid pixels map exactly to -1, ensuring a distinct representation for missing measurements within the normalized domain. Unlike Marigold and E2E-FT (who normalize only dense targets) we apply the same per-image normalization to both dense and sparse depth maps so their dynamic ranges remain aligned throughout training. RGB is scaled to $[-1, 1]$ with the same convention. Depth tensors (dense and sparse) are replicated to three channels to match the frozen VAE encoder; spatial transforms (resize/crop/flip) are applied consistently across RGB, dense, and sparse inputs. The per-image affine normalization removes global scale and offset so that all samples are represented in a consistent, dimensionless range. This makes the model invariant to scene scale and camera calibration differences, encouraging it to learn relative depth structure rather than absolute metric depth. Normalizing to the Stable Diffusion VAE's expected $[-1, 1]$ range also ensures that input magnitudes remain within the activation bounds of the pretrained encoder, preventing saturation and instability during training. Assigning a fixed value of -1 to invalid pixels provides a distinct representation for missing regions, enabling the model to recognize and inpaint them effectively.

To balance indoor and outdoor scene distributions, we adopt the mixed-dataset training strategy used in prior LDM-based depth estimation work. Each batch is sampled with a fixed 9:1 ratio of Hypersim to VKITTI2 using the `MixedDataLoader`. This configuration prevents overfitting to indoor spatial layouts and lighting statistics while maintaining sufficient exposure to outdoor driving scenes, which are more geometrically expansive but less textured. By mixing datasets with complementary depth ranges and visual domains, the model learns features that generalize across different scales, camera baselines, and scene types. The data loading and preprocessing routines implementing this scheme are provided in `load_metric.py`.

4.2.2 Fine-tuning procedure

Setup. We fine-tune a ControlNet on top of `SD-E2E-FT-depth` while keeping the Stable Diffusion backbone frozen. The VAE, text encoder, and U-Net weights remain fixed and are set to evaluation mode, such that only the ControlNet parameters are optimized. To accommodate the concatenated RGB and depth latents, the U-Net input layer is expanded from 4 to 8 channels by duplicating the pretrained convolutional kernel along the channel dimension and scaling its weights by $\frac{1}{2}$ to maintain stable activation magnitudes. We enable memory-efficient attention (`xFormers`) and gradient checkpointing to reduce GPU memory usage during fine-tuning.

Conditioning and inputs. Let $\mathcal{E}$ and $\mathcal{D}$ denote the frozen Stable Diffusion VAE encoder and decoder, respectively. Given an RGB image $\mathbf{i}_{\text{RGB}} \in \mathbb{R}^{H\times W\times 3}$ and its corresponding dense depth map $\mathbf{d}_{\text{dense}} \in \mathbb{R}^{H\times W}$, both preprocessed, we compute

$$\mathbf{x} = \mathcal{E}(\mathbf{i}_{\text{RGB}}) \in \mathbb{R}^{4\times h\times w}, \qquad \mathbf{z}_0^{(\text{init})} = \mathbf{0} \in \mathbb{R}^{4\times h\times w}.$$

These are concatenated along the channel dimension to form the U-Net input:

$$\mathbf{c} = [\,\mathbf{x},\ \mathbf{z}_0^{(\text{init})}\,] \in \mathbb{R}^{8\times h\times w}.$$

The sparse depth image $\mathbf{d}_{\text{sparse}}$, normalized and replicated to three channels, serves as the ControlNet's conditioning input. Its lightweight convolutional encoder extracts feature maps that are injected into the U-Net's down and mid blocks following the standard ControlNet design. We use an empty text embedding as in E2E-FT [20].

Single-step training (fixed $t{=}T$). Unlike Marigold [43], which samples random timesteps, we follow E2E-FT [20] and fix the timestep to $t{=}T$ to enable deterministic, single-step prediction. The denoiser predicts a v-parameterized target

$$\hat{\mathbf{v}}_T = \mathbf{v}_\theta(\mathbf{c}, T),$$

which is then converted into the clean latent depth estimate

$$\hat{\mathbf{z}}_0 = \sqrt{\bar{\alpha}_T}\,\mathbf{z}_T - \sqrt{1-\bar{\alpha}_T}\,\hat{\mathbf{v}}_T.$$

Since $\mathbf{z}_T = \mathbf{0}$ at initialization, this simplifies to

$$\hat{\mathbf{z}}_0 = -\sqrt{1-\bar{\alpha}_T}\,\hat{\mathbf{v}}_T,$$

meaning the model directly predicts the clean depth latent in a single step up to a constant factor. The predicted depth map is then obtained by decoding and averaging over the latent channels:

$$\hat{\mathbf{d}} = \text{mean}\big(\mathcal{D}(\hat{\mathbf{z}}_0),\ \text{channels}\big) \in \mathbb{R}^{H\times W}.$$

The VAE remains frozen, but gradients are propagated through it to optimize the ControlNet pathway.

Loss (affine-invariant). The training objective follows the scale-and-shift invariant (SSI) loss introduced in E2E-FT [20]. Let $\hat{\mathbf{d}} \in \mathbb{R}^{H\times W}$ denote the predicted dense depth map decoded from the model output, and let $\mathbf{d}^* \in \mathbb{R}^{H\times W}$ denote the corresponding dense ground-truth depth map. A binary validity mask $\mathbf{M}{\in}\{0,1\}^{H\times W}$ is used to identify valid pixels, where $\mathbf{M}(i,j)=1$ if $\mathbf{d}^*(i,j)$ lies within the dataset's valid depth range $[d_{\min}, d_{\max}]$, and $\mathbf{M}(i,j)=0$ otherwise. The SSI formulation estimates a global scale factor $s{\in}\mathbb{R}$ and shift $t{\in}\mathbb{R}$ that best align the prediction to the ground truth over valid pixels by solving

$$(s,t) = \arg\min_{s,t} \big\|(s\,\hat{\mathbf{d}} + t) - \mathbf{d}^*\big\|_{1,\mathbf{M}},$$

where $\|\cdot\|_{1,\mathbf{M}}$ denotes the masked L_1 norm computed only over pixels with $\mathbf{M}(i,j)=1$. The final SSI loss is then defined as

$$\mathcal{L}_{\text{SSI}} = \frac{1}{|\mathbf{M}|}\,\big\|(s\,\hat{\mathbf{d}} + t) - \mathbf{d}^*\big\|_{1,\mathbf{M}},$$

where $|\mathbf{M}| = \sum_{i,j}\mathbf{M}(i,j)$ is the number of valid pixels. This affine normalization removes the influence of global scale and offset differences between $\hat{\mathbf{d}}$ and $\mathbf{d}^*$, allowing the model to learn relative depth structure rather than absolute metric scale.

Zero-latent initialization. Initializing the depth branch with $\mathbf{z}_T = \mathbf{0}$ matches the mean of the diffusion prior, avoids injecting stochastic noise during training, and yields a deterministic mapping from $[\mathbf{x}, \mathbf{0}]$ to $\hat{\mathbf{z}}_0$. Empirically, this stabilizes optimization and prevents the ControlNet from learning to compensate for random noise patterns, thereby protecting the frozen U-Net weights from undesired gradients.

Optimization and schedule. Training is performed for 20,000 iterations using a single A100 80GB gpu and the AdamW optimizer [55] with a base learning rate of 3×10^{-5}. The iteration-wise exponential scheduler decays the learning rate to 0.01 of its initial value by the final step, after a 100-step warm-up period. We use a batch size of 1 with gradient accumulation over 2 steps (effective batch size of 2), `num_workers= 0`, and gradient clipping applied only to the trainable ControlNet parameters. Unless otherwise stated, other hyperparameters follow Marigold and E2E-FT.

Noise schedulers. Training employs the DDPM scheduler, while validation and inference use the DDIM scheduler [84]. Regardless of the noise scheduler, we use `timestep_spacing='trailing'` to ensure timestep–noise alignment for single-step prediction, as recommended by E2E-FT. The ControlNet conditioning scale is set to the default value of 1.0 during training for all experiments.

Validation protocol. After a fixed number of training steps, we perform qualitative validation to monitor model progress. At each validation interval, we run inference on a small, fixed subset of RGB–sparse depth pairs sampled from the Hypersim and VKITTI2 validation splits. During validation, the frozen `StableDiffusionControlNetPipeline` is instantiated with the current ControlNet weights and executed using the DDIM scheduler [84] for deterministic single-step prediction. This differs from training, which uses the DDPM scheduler for stochastic noise injection. For each validation sample, we load and normalize the sparse depth map and corresponding RGB image, apply the same preprocessing pipeline as in training, and perform a single forward pass to produce the predicted dense depth. The outputs are saved as color-mapped depth images for visual inspection.

No quantitative metrics are computed at this stage; instead, validation serves to visually confirm that the ControlNet learns to recover fine-grained scene geometry and does not diverge during training. The saved predictions are logged to TensorBoard or to Weights & Biases for tracking over time. This qualitative check is particularly useful in diffusion-based depth models, where global consistency and smooth depth transitions are not always reflected by scalar losses.

4.3 Inference with Naïve ControlNet Model

Given a preprocessed RGB image $\mathbf{i}_{\text{RGB}} \in \mathbb{R}^{H \times W \times 3}$ and its corresponding sparse depth map $\mathbf{d}_{\text{sparse}} \in \mathbb{R}^{H \times W}$ (triplicated to three channels), inference is performed in a single denoising step using the trailing timestep schedule.

Encoding and initialization. The RGB image is encoded by the frozen Stable Diffusion VAE encoder $\mathcal{E}(\cdot)$ to obtain the RGB latent

$$\mathbf{x} = \mathcal{E}(\mathbf{i}_{\text{RGB}}) \in \mathbb{R}^{4 \times h \times w},$$

where (h, w) denote the latent spatial resolution. The predicted depth latent is initialized with zeros,

$$\mathbf{z}_T = \mathbf{0} \in \mathbb{R}^{4 \times h \times w},$$

and concatenated with the RGB latent to form the U-Net input

$$\mathbf{u}_T = [\, \mathbf{x},\, \mathbf{z}_T \,] \in \mathbb{R}^{8 \times h \times w}.$$

ControlNet conditioning. The sparse depth image (three channels after triplication) is provided to the ControlNet encoder $\mathcal{E}_{\text{ctrl}}(\cdot)$, producing a latent-space feature embedding

$$\mathbf{e}_{\text{ctrl}} = \mathcal{E}_{\text{ctrl}}(\mathbf{d}_{\text{sparse}}) \in \mathbb{R}^{C\times h\times w}.$$

Let $\text{ControlNet}_{\psi}(\cdot)$ denote the full ControlNet module with trainable parameters ψ. Following Zhang *et al.* [106], each ControlNet block mirrors the structure of the corresponding down-sampling or middle block of the frozen U-Net. If a U-Net block transforms a feature map $\mathbf{f}_{\ell}^{\downarrow}$ into $\mathbf{f}_{\ell+1}^{\downarrow} = F_{\ell}(\mathbf{f}_{\ell}^{\downarrow};\theta_{\ell})$, the ControlNet introduces a parallel trainable branch defined as

$$\mathbf{f}_{\ell,c}^{\downarrow} = F_{\ell}(\mathbf{f}_{\ell}^{\downarrow};\theta_{\ell}) + Z\Big(F_{\ell,c}\big(\mathbf{f}_{\ell}^{\downarrow} + Z(\mathbf{e}_{\text{ctrl}};\theta_{\ell,z1});\theta_{\ell,c}\big);\theta_{\ell,z2}\Big),$$

where $F_{\ell,c}$ is the trainable ControlNet copy of the block and $Z(\cdot;\theta_{\ell,zi})$ are zero-initialized 1×1 convolutions that gate the effect of the conditioning feature $\mathbf{e}_{\text{ctrl}}$.

The ControlNet produces additive residual feature maps that are injected into the frozen U-Net. We denote by $\Delta\mathbf{f}_{\ell}^{\downarrow}$ the residual feature map added to the ℓ-th down-sampling block and by $\Delta\mathbf{f}^{\text{mid}}$ the residual for the middle block. The collection of all down-block residuals is written as $\{\Delta\mathbf{f}_{\ell}^{\downarrow}\}_{\ell}$, where the curly brackets indicate the set of residuals across all encoder layers.

During inference, with the trailing timestep t_T (single-step prediction) and an empty text embedding, the ControlNet produces these residuals as

$$\{\Delta\mathbf{f}_{\ell}^{\downarrow}\}_{\ell},\ \Delta\mathbf{f}^{\text{mid}} = \text{ControlNet}_{\psi}(\mathbf{u}_T,\ t_T,\ \mathbf{e}_{\text{ctrl}};\ s_{\text{ctrl}}),$$

where s_{ctrl} is the `controlnet_conditioning_scale` that determines the strength of the conditioning signal. These residuals are then added to the corresponding U-Net feature maps before decoding.

Denoising and decoding. Using the trailing timestep t_T and an empty text embedding, the ControlNet injects the residual feature maps $\{\Delta\mathbf{f}_{\ell}^{\downarrow}\}_{\ell}$ and $\Delta\mathbf{f}^{\text{mid}}$ into the corresponding down-sampling and middle blocks of the frozen E2E-FT U-Net. The U-Net then predicts a residual latent representation

$$\mathbf{r}_{\theta} = f_{\theta}\Big(\mathbf{u}_T,\ t_T;\ \{\Delta\mathbf{f}_{\ell}^{\downarrow}\}_{\ell},\ \Delta\mathbf{f}^{\text{mid}}\Big) \in \mathbb{R}^{4\times h\times w},$$

where $f_{\theta}(\cdot)$ denotes the U-Net denoiser parameterized by frozen weights θ. The output $\mathbf{r}_{\theta}$ corresponds to the v-parameterized residual under the DDIM formulation. A single scheduler update then maps the noisy latent $\mathbf{z}_T$ and predicted residual $\mathbf{r}_{\theta}$ to an estimate of the clean latent depth:

$$\hat{\mathbf{z}}_0 = \Phi_{\text{DDIM}}(\mathbf{z}_T,\ \mathbf{r}_{\theta},\ t_T)\,,$$

where $\Phi_{\text{DDIM}}(\cdot)$ denotes the deterministic one-step DDIM update with `timestep_spacing='trailing'` to ensure proper timestep–noise alignment.

The clean latent $\hat{\mathbf{z}}_0$ is decoded using the frozen VAE decoder $\mathcal{D}(\cdot)$, scaled by the Stable Diffusion latent factor s_{VAE} (typically s_{VAE}=0.18215), to produce a three-channel normalized depth image:

$$\hat{\mathbf{y}} = \mathcal{D}\left(\frac{\hat{\mathbf{z}}_0}{s_{\text{VAE}}}\right) \in \mathbb{R}^{3\times h\times w}.$$

A single-channel dense depth prediction is then obtained by averaging across channels:

$$\hat{\mathbf{d}} = \text{mean}_{\text{ch}}(\hat{\mathbf{y}}) \in \mathbb{R}^{1\times h\times w}.$$

Here, $\mathcal{D}(\cdot)$ denotes the frozen VAE decoder, $\hat{\mathbf{z}}_0$ is the clean latent estimate, and $\hat{\mathbf{y}}$ represents the reconstructed three-channel depth image in the normalized $[-1, 1]$ range. The operator $\mathrm{mean}_{\mathrm{ch}}(\cdot)$ averages the three output channels to yield the single-channel dense depth map $\hat{\mathbf{d}}$. Finally, $\hat{\mathbf{d}}$ is clamped to $[-1, 1]$, linearly normalized to $[0, 1]$ per image, and resized to the input resolution (H, W). At inference, the control strength is set by s_{ctrl}; all reported results use s_{ctrl}=1.0 and a single DDIM inference step.

Notes on the implementation. (i) The ControlNet receives a 3-channel control image and produces down/mid residuals that are added into the UNet (no architectural changes beyond conditioning). (ii) The UNet input is $[\,\mathbf{x}, \mathbf{z}_T\,]$ with $\mathbf{z}_T$=$\mathbf{0}$; the text conditioning is the empty embedding. (iii) Single-step DDIM with trailing spacing is used for deterministic inference. (iv) All VAE modules remain frozen; decoding applies the inverse scaling factor before the forward pass of $\mathcal{D}(\cdot)$.

4.3.1 Limitations of the Naïve ControlNet Model

Despite attaching a ControlNet to `SD-E2E-FT-depth`, the naïve variant delivers only marginal gains after 20k steps relative to the untrained baseline. In practice, predictions change little across indoor and outdoor benchmarks, indicating that the model does not reliably exploit the sparse conditioning.

Concretely, let $\mathcal{E}$ be the frozen SD VAE encoder and $\mathbf{x} = \mathcal{E}(\mathbf{i}_{\mathrm{RGB}}) \in \mathbb{R}^{4\times h\times w}$; E2E-FT initializes the depth latent with zeros, $\mathbf{z}_T = \mathbf{0}$, and feeds the UNet the concatenation $\mathbf{u}_T = [\mathbf{x}, \mathbf{0}]$. We refer to the resulting pretrained mapping $[\mathbf{x}, \mathbf{0}] \xrightarrow{\text{frozen UNet}} \hat{\mathbf{z}}_0 \xrightarrow{\text{frozen VAE}} \hat{\mathbf{d}}$ as the "$[\mathbf{x}, \mathbf{0}]$ path." The "controller" is the ControlNet branch that encodes the sparse-depth control image and injects *additive* residual feature maps into the UNet's down and mid blocks; these residuals are zero-initialized (zero-conv) and scaled by the user-set `controlnet_conditioning_scale`.

In our setup the backbone (UNet and VAE) is frozen and already produces a strong one-step prediction from $[\mathbf{x}, \mathbf{0}]$, while the training objective is a scale-and-shift invariant (SSI) loss against dense ground truth with *no* explicit penalty for mismatching the observed sparse pixels. Under this objective, the optimizer can reduce the loss almost entirely via the $[\mathbf{x}, \mathbf{0}]$ path, so gradients flowing into the controller remain small and its residuals stay near zero.

Two additional factors further limit the controller's effect: (i) the control image is provided only in pixel space without an explicit binary validity mask, forcing the controller to infer missingness from depth values (e.g., a sentinel -1) rather than receiving a clean measured-vs-missing gate; and (ii) conditioning occurs in pixel space while prediction is produced in the VAE latent domain, so the injected residuals are not naturally aligned with the features that determine the output.

Finally, portions of the synthetic training regime contain relatively high sparse-point counts; however, because the loss never *requires* exact agreement at those measured locations, additional points do not increase the incentive to use the controller, and the RGB-only baseline already reconstructs plausible geometry in many such frames, leaving little marginal error for guidance to reduce. Combined with a modest effective batch size in single-step training, these choices make the controller neither compulsory (no sparse-fidelity term) nor well-signaled (no explicit mask), and it remains largely inactive; performance is therefore driven by the underlying E2E-FT checkpoint.

These observations motivate an inpainting-style variant that (i) supplies an explicit validity mask, (ii) aligns conditioning with the latent prediction domain, and (iii) augments the objective with a sparse-consistency term; we describe this design next.

4.4 Training the ControlNet E2E FT Inpainting Model for Guided Monocular Depth Estimation

The training configuration follows the setup described in Section 4.2. Training is performed in the single-step regime using the trailing timestep, with the same optimizer, learning-rate schedule, and mixed-dataset sampling strategy as in the naïve variant. The Stable Diffusion VAE, text encoder, and U-Net remain frozen, with only the ControlNet parameters optimized; all other components such as the optimizer, noise scheduler, and data loading configuration follow the setup described in Section 4.2.

Inputs and conditioning. Let $\mathcal{E}$ and $\mathcal{D}$ denote the frozen Stable Diffusion VAE encoder and decoder. Given an RGB image $\mathbf{i}_{\mathrm{RGB}} \in \mathbb{R}^{H\times W\times 3}$ and its corresponding sparse depth map $\mathbf{d}_{\mathrm{sparse}} \in \mathbb{R}^{H\times W}$ (both preprocessed as described in Section 4.2.1), we compute the encoded RGB latent

$$\mathbf{x} = \mathcal{E}(\mathbf{i}_{\mathrm{RGB}}) \in \mathbb{R}^{4\times h\times w}$$

and the encoded sparse-depth latent

$$\mathbf{z}_c = \mathcal{E}(\mathbf{d}_{\mathrm{sparse}}) \in \mathbb{R}^{4\times h\times w}.$$

The binary validity mask $\mathbf{m} \in \{0,1\}^{H\times W}$ is defined over pixel coordinates (i,j) as

$$\mathbf{m}(i,j) = \begin{cases} 1, & \text{if } \mathbf{d}_{\mathrm{sparse}}(i,j) \in [d_{\min},\, d_{\max}], \\ 0, & \text{otherwise.} \end{cases}$$

Here, $\mathbf{d}_{\mathrm{sparse}}(i,j)$ denotes the sparse depth value at pixel coordinate (i,j), and $d_{\min}$ and $d_{\max}$ represent the dataset-specific near and far depth bounds used during normalization (e.g., $d_{\min} = 10^{-5}$ and $d_{\max} = 65\,\mathrm{m}$ for Hypersim, $d_{\max} = 80\,\mathrm{m}$ for VKITTI2). Pixels whose sparse depth values are zero, negative, or outside this range are considered invalid and assigned a mask value of 0. The mask is then downsampled to latent resolution (h,w) via nearest-neighbour interpolation to obtain $\mathbf{m}_{\downarrow} \in \{0,1\}^{1\times h\times w}$, preserving the binary validity pattern of the sparse input after downsampling.

The U-Net input remains identical to the naïve variant:

$$\mathbf{u}_T = [\,\mathbf{x},\, \mathbf{z}_T\,] \in \mathbb{R}^{8\times h\times w}, \qquad \mathbf{z}_T = \mathbf{0},$$

where $\mathbf{z}_T$ denotes the zero-initialized predicted-depth latent at timestep T. The ControlNet, however, now receives a 9-channel latent-space condition tensor

$$\mathbf{c} = [\,\mathbf{x},\, \mathbf{z}_c,\, \mathbf{m}_{\downarrow}\,] \in \mathbb{R}^{9\times h\times w}.$$

This concatenation is constructed directly before the ControlNet forward pass. Unlike the naïve variant, which provided only the 3-channel sparse depth image in pixel space, this conditioning operates entirely in the latent domain, co-registered with the U-Net features and explicitly augmented with a binary mask.

Latent-resolution ControlNet embedding. To maintain spatial alignment, all convolutions in the ControlNet's conditional embedding path originally defined with stride $(2,2)$ are replaced by $(1,1)$. This modification ensures that the conditional embedding preserves the latent resolution (h,w), preventing unwanted downsampling and maintaining a clear transition between valid and missing regions defined by $\mathbf{m}_{\downarrow}$.

Forward pass. Using the trailing timestep t_T and an empty text embedding, the ControlNet processes $\mathbf{c}$ and produces additive residuals that are injected into the corresponding down and mid blocks of the U-Net with a conditioning scale of s_{ctrl}=1.0. The U-Net then predicts a residual latent representation $\mathbf{r}_\theta \in \mathbb{R}^{4\times h\times w}$ under the v-parameterization. Given $\mathbf{z}_T$=$\mathbf{0}$, the clean latent estimate is computed as

$$\hat{\mathbf{z}}_0 = -\sqrt{1-\bar{\alpha}_T}\,\mathbf{r}_\theta.$$

Decoding with the frozen VAE and averaging across channels yields the normalized dense depth prediction:

$$\hat{\mathbf{y}} = \mathcal{D}\left(\frac{\hat{\mathbf{z}}_0}{s_{\text{VAE}}}\right) \in \mathbb{R}^{3\times h\times w}, \qquad \hat{\mathbf{d}} = \text{mean}_{\text{ch}}(\hat{\mathbf{y}}) \in \mathbb{R}^{1\times h\times w}.$$

Here, $\mathcal{D}(\cdot)$ denotes the frozen VAE decoder, $\hat{\mathbf{z}}_0$ is the estimated clean latent produced by the denoiser, and s_{VAE} is the latent scaling factor used in Stable Diffusion (typically s_{VAE}=0.18215) that converts between pixel-space values and latent magnitudes. The decoded tensor $\hat{\mathbf{y}}$ represents the reconstructed three-channel depth image in the normalized $[-1, 1]$ range, while $\text{mean}_{\text{ch}}(\cdot)$ computes the mean across the three decoder output channels to obtain the single-channel dense depth map $\hat{\mathbf{d}}$.

Loss functions. The total training objective combines the affine-invariant (SSI) depth loss with a sparse-consistency term that enforces agreement between the predicted dense depth and the input sparse measurements:

$$\mathcal{L} = \lambda_{\text{main}}\,\mathcal{L}_{\text{SSI}}(\hat{\mathbf{d}}, \mathbf{d}^*; \mathbf{M}) + \lambda_{\text{sp}}\,\mathcal{L}_{\text{sparse}}(\hat{\mathbf{y}}, \mathbf{d}_{\text{sparse}}; \mathbf{m}).$$

where $\lambda_{\text{main}}, \lambda_{\text{sp}} > 0$ are scalar weights that control the relative influence of the two losses.

The variable $\hat{\mathbf{d}} \in \mathbb{R}^{1\times h\times w}$ denotes the predicted dense depth map decoded from the latent output $\hat{\mathbf{z}}_0$, and $\mathbf{d}^* \in \mathbb{R}^{H\times W}$ is the corresponding dense ground-truth depth map. $\hat{\mathbf{y}} \in \mathbb{R}^{3\times h\times w}$ represents the three-channel decoded prediction before channel averaging (see previous section), while $\mathbf{d}_{\text{sparse}} \in \mathbb{R}^{H\times W}$ is the sparse input depth image, and $\mathbf{m} \in \{0,1\}^{H\times W}$ is its binary validity mask. $\mathbf{M} \in \{0,1\}^{H\times W}$ denotes the ground-truth validity mask used to identify valid dense pixels (i.e., points within the dataset's near–far range).

The first term, $\mathcal{L}_{\text{SSI}}$, is the same scale-and-shift invariant loss from the naïve variant (see Section 4.2 for more details). The second term, $\mathcal{L}_{\text{sparse}}$, enforces pointwise consistency between the predicted depth map and input sparse depths at all valid sparse locations:

$$\mathcal{L}_{\text{sparse}} = \frac{1}{\|\mathbf{m}\|_1} \sum_{(i,j):\mathbf{m}(i,j)=1} \rho(\hat{\mathbf{y}}_1(i,j) - \mathbf{d}_{\text{sparse},1}(i,j))$$

where $\rho(\cdot)$ is a robust penalty function, specifically the Huber loss, although ℓ_1 and ℓ_2 variants are also supported. The subscript 1 refers to the first channel of each three-channel depth tensor. The denominator $\|\mathbf{m}\|_1$ normalizes the loss by the number of valid sparse points.

The sparse-consistency loss introduces a direct supervision signal that links the ControlNet input to the predicted output at measured pixels. The scale-and-shift invariant loss $\mathcal{L}_{\text{SSI}}$ evaluates predictions only up to a global scale and offset, meaning that a depth map can be uniformly rescaled or shifted without increasing the loss. The sparse-consistency term counteracts this property by encouraging the model to produce predictions whose depth values at pixels with valid sparse measurements are numerically close to those input values. This term guides the optimization process so that the ControlNet contributes meaningful local adjustments in regions where sparse depth information is available, rather than allowing the pretrained $[\mathbf{x}, \mathbf{0}]$ pathway to dominate the loss minimization.

Summary. Compared to the naïve variant, this design introduces three concrete modifications. First, the inclusion of an explicit binary mask $\mathbf{m}$ provides a clear separation between measured and unmeasured regions for the ControlNet, allowing its residuals to be applied selectively rather than uniformly across the latent feature map. Second, the conditioning inputs $\mathbf{x}$ and $\mathbf{z}_c$ are defined in the same latent space and at the same spatial resolution as the U-Net input, ensuring that the ControlNet operates on features directly relevant to depth generation. Third, the additional sparse-consistency term supplements the scale-and-shift invariant loss by encouraging the model to produce predicted depth values that are numerically closer to the provided sparse measurements at valid pixel locations, even if exact equality is not guaranteed. Throughout training, the Stable Diffusion VAE remains frozen so that the learned latent representation and its decoder retain their pretrained structure. This ensures that the decoded predictions preserve the semantic integrity of the latent depth representation while adapting to better match the sparse input. Together, these changes make the sparse conditioning both functionally necessary and effective during optimization, leading to more consistent and spatially aligned guided depth predictions than in the naïve formulation.

4.5 Inference with the ControlNet Inpainting Model

Inference largely mirrors the naïve variant in Section 4.3; we only detail the parts that differ.

Inputs and encoding. Given a preprocessed RGB image $\mathbf{i}_{\mathrm{RGB}} \in \mathbb{R}^{H\times W\times 3}$ and sparse depth $\mathbf{d}_{\mathrm{sparse}} \in \mathbb{R}^{H\times W}$, the frozen VAE encoder $\mathcal{E}$ produces

$$\mathbf{x} = \mathcal{E}(\mathbf{i}_{\mathrm{RGB}}) \in \mathbb{R}^{4\times h\times w}, \qquad \mathbf{z}_c = \mathcal{E}(\mathbf{d}_{\mathrm{sparse}}) \in \mathbb{R}^{4\times h\times w}.$$

The predicted-depth latent is initialized to zero, $\mathbf{z}_T = \mathbf{0} \in \mathbb{R}^{4\times h\times w}$, and the U-Net input is $\mathbf{u}_T = [\,\mathbf{x}, \mathbf{z}_T\,] \in \mathbb{R}^{8\times h\times w}$. The binary validity mask $\mathbf{m} \in \{0,1\}^{H\times W}$ (defined as in Section 4.4) is downsampled by nearest neighbour to latent resolution to obtain $\mathbf{m}_{\downarrow} \in \{0,1\}^{1\times h\times w}$. The ControlNet condition is the latent-space concatenation

$$\mathbf{c} = [\,\mathbf{x}, \mathbf{z}_c, \mathbf{m}_{\downarrow}\,] \in \mathbb{R}^{9\times h\times w}.$$

ControlNet application and denoising. With trailing timestep t_T for single-step prediction and an empty text embedding, the ControlNet produces additive residuals for the U-Net encoder (down blocks) and middle block,

$$\left\{\Delta\mathbf{f}_\ell^{\downarrow}\right\}_\ell,\ \Delta\mathbf{f}^{\mathrm{mid}} = \mathrm{ControlNet}_\psi(\mathbf{u}_T,\ t_T,\ \mathbf{c};\ s_{\mathrm{ctrl}})\,,$$

where ψ are ControlNet parameters and s_{ctrl} is the control strength (we use s_{ctrl}=1.0). Injecting these residuals, the frozen U-Net f_θ predicts the v-parameterized residual

$$\mathbf{r}_\theta = f_\theta\Big(\mathbf{u}_T,\ t_T;\ \left\{\Delta\mathbf{f}_\ell^{\downarrow}\right\}_\ell,\ \Delta\mathbf{f}^{\mathrm{mid}}\Big) \in \mathbb{R}^{4\times h\times w}.$$

With $\mathbf{z}_T = \mathbf{0}$ and $\bar{\alpha}_T$ the cumulative noise coefficient at t_T, the clean latent estimate is

$$\hat{\mathbf{z}}_0 \;=\; -\sqrt{1-\bar{\alpha}_T}\;\mathbf{r}_\theta \;\in \mathbb{R}^{4\times h\times w}.$$

Decoding and post-processing. The frozen decoder $\mathcal{D}$ maps the latent to pixel space using the Stable Diffusion scaling factor s_{VAE},

$$\hat{\mathbf{y}} = \mathcal{D}\left(\frac{\hat{\mathbf{z}}_0}{s_{\mathrm{VAE}}}\right) \in \mathbb{R}^{3\times h\times w}, \qquad \hat{\mathbf{d}} = \mathrm{mean}_{\mathrm{ch}}(\hat{\mathbf{y}}) \in \mathbb{R}^{1\times h\times w}.$$

We clamp $\hat{\mathbf{d}}$ to $[-1, 1]$, apply a per-image linear min–max normalization to $[0, 1]$, and resize to (H, W) to match the input resolution (identical to the naïve variant's post-processing).

Relative to Section 4.3, the only changes are: (i) ControlNet conditioning is performed in latent space with $\mathbf{c} = [\mathbf{x}, \mathbf{z}_c, \mathbf{m}_\downarrow]$ at resolution (h, w); and (ii) the explicit mask $\mathbf{m}_\downarrow$ gates guidance to measured regions. All other components (single-step trailing schedule, v-parameterization, decoding and normalization) are unchanged.

Chapter 5

Experiments

5.1 Evaluation datasets

5.1.1 Experimental Setup

Datasets. We evaluate our guided monocular depth estimation models together with existing monocular depth estimation and depth completion methods on the same KITTI [21], ScanNet [15], DIODE [91], and NYUv2 [82] test splits used by Marigold [43] and E2E-FT [20]. Using identical datasets for both monocular and guided variants provides a controlled comparison of how introducing sparse depth conditioning influences prediction accuracy under the same image–depth distributions, without the confounding effects of differing data sources or evaluation protocols. Images are used at their native resolutions: KITTI at 1216×352 with the benchmark crop, NYUv2 and ScanNet at 640×480, and DIODE at 1024×768. These settings are consistent with prior work and the dataset documentation.

For guided inference, we precompute sparse depth maps by uniformly sampling K valid ground-truth points per image. Let $\mathbf{d}^* \in \mathbb{R}^{H\times W}$ denote the dense ground-truth depth, and let $[d_{\min}, d_{\max}]$ be the dataset-specific valid range. Pixels (i,j) satisfying $\mathbf{d}^*(i,j) \in [d_{\min}, d_{\max}]$ are eligible for sampling; a subset S of size K is drawn uniformly without replacement, and the sparse map is defined as

$$\mathbf{d}_{\text{sparse}}(i,j) = \begin{cases} \mathbf{d}^*(i,j), & (i,j) \in S, \\ 0, & \text{otherwise.} \end{cases}$$

Sparse maps are generated once and reused for all models. We sample K=80,000 points for all datasets unless stated otherwise. The sparse maps are provided to our methods as the conditional input, as well as to Marigold-DC. Depths are stored following the native formats: NYUv2/ScanNet as 16-bit PNGs scaled by $1/1000$, KITTI as 16-bit PNGs scaled by $1/256$, and DIODE as `.npy` arrays in meters. Random seeds are fixed to ensure reproducibility.

Metrics. We report the standard metrics used in monocular and sparse-guided depth estimation: AbsRel, SqRel, RMSE, $\text{RMSE}_{\log}$, $\log_{10}$, δ_t (t=1, 2, 3), iRMSE, and SI-Log. All metrics are computed on valid pixels after per-image scale and shift alignment:

$$(s,t) = \arg\min_{s,t} \sum_{p\in\Omega} (s\,\hat{d}(p) + t - d^*(p))^2, \qquad \hat{d}_{\text{aligned}} = s\,\hat{d} + t,$$

where Ω denotes the validity mask. For KITTI and VKITTI, evaluation is performed on the 1216×352 benchmark crop and the Eigen split; for NYUv2 we use the standard Eigen crop $[45{:}471]\times[41{:}601]$. DIODE and ScanNet use their full valid regions.

Implementation details. Inference uses the `StableDiffusionControlNetInpaintPipeline` with a single deterministic DDIM step (`timestep_spacing=trailing`), zero-latent initialization, and an empty text prompt. The ControlNet conditioning scale is fixed to s_{ctrl}=1.0. GPU offloading and xFormers attention are enabled for memory efficiency. Sparse control inputs are single-channel tensors normalized to $[0, 1]$ with an ϵ=0.02 margin at valid pixels, passed as $(1, 1, H, W)$ to the pipeline. Predictions are stored as `.npy` float arrays and evaluated with the metrics above. The following validity ranges are used:

$$[d_{\min}, d_{\max}] = \begin{cases} [10^{-3}, 10.0]\ \text{m}, & \text{NYUv2, ScanNet,} \\ [10^{-5}, 80.0]\ \text{m}, & \text{KITTI,} \\ [0.6, 350.0]\ \text{m}, & \text{DIODE.} \end{cases}$$

All experiments use a single forward pass per image with no test-time optimization.

—

5.2 Results

Comparisons with monocular depth estimation (MDE) methods. Table 5.1 presents a quantitative comparison between our models and leading monocular depth estimators. The naïve ControlNet variant already matches or exceeds Marigold and E2E-FT on most indoor datasets, confirming that sparse guidance can improve latent diffusion depth prediction even without mask conditioning. The inpainting-based ControlNet further improves upon the naïve variant across all datasets except KITTI, achieving lower AbsRel and higher δ_1 accuracy. These results demonstrate that the proposed modifications—latent-domain conditioning, explicit validity masks, and sparse-consistency loss—enable the model to exploit sparse geometric cues effectively.

On DIODE, our ControlNet E2E-FT achieves a 0.239 AbsRel and 0.808 δ_1, outperforming all diffusion-based MDE methods. On NYUv2 and ScanNet, it attains the best or second-best results among all LDM-based approaches, showing consistent indoor performance. The KITTI results are slightly weaker, likely due to limited outdoor variation in the training data and the narrow baseline of the depth maps. As expected, Metric3D variants outperform diffusion-based estimators overall, reflecting their supervised training and architectural specialization for depth completion. Nonetheless, our model remains competitive despite using a single forward pass and no depth supervision.

Comparison with depth completion models. Although our method surpasses monocular estimators, it does not yet outperform dedicated depth completion methods such as Marigold-DC, which perform iterative test-time optimization. This difference highlights the efficiency–accuracy trade-off: our approach achieves comparable structural accuracy using only deterministic feed-forward inference, whereas completion models rely on adaptive optimization per image. Our method therefore provides a faster, inference-only alternative suitable for applications requiring consistent, real-time predictions.

Overall, the ControlNet Inpainting E2E FT Model improves upon the baseline diffusion depth estimators across all indoor datasets and narrows the gap to depth completion methods, confirming that explicit sparse conditioning can enhance latent diffusion performance without sacrificing efficiency.

Further experimental details, including extended quantitative comparisons and results from the naive model with varying sparse point densities and checkpoints, are provided in Appendix A.1 for completeness.

Table 5.1: **Quantitative comparison** of Marigold and other state-of-the-art affine-invariant depth estimators on several zero-shot benchmarks. All metrics† are presented in percentage terms; **bold** numbers denote the best performance and underlined numbers denote the second best amongst non-Marigold-DC frameworks. *Ours (Naive Implementation)* refers to the baseline described in §4.2, while *Ours (ControlNet E2E FT)* corresponds to the improved model introduced in §4.4. Our method achieves consistently strong results across indoor and outdoor benchmarks without training on real depth data.

Method	DIODE		KITTI		NYUv2		ScanNet	
	AbsRel↓	δ_1 ↑	AbsRel↓	δ_1 ↑	AbsRel↓	δ_1 ↑	AbsRel↓	δ_1 ↑
Marigold E2E FT	0.302	0.779	0.096	0.919	0.052	0.966	0.096	0.966
Marigold	0.302	0.776	0.099	0.916	0.055	0.964	0.099	0.964
Lotus-D	**0.228**	0.738	0.081	0.931	0.051	**0.972**	0.081	**0.972**
Lotus-G	0.230	0.729	0.085	0.922	0.054	0.967	0.085	0.967
SD E2E FT	0.303	0.777	0.096	0.921	0.054	0.965	0.096	0.965
Depth Anything	0.371	0.668	0.309	0.454	0.117	0.873	0.309	0.873
Depth Anything v2	0.373	0.665	0.307	0.455	0.121	0.865	0.307	0.865
Metric3D v1 large	0.446	0.750	**0.058**	**0.964**	0.066	0.948	**0.058**	0.948
Marigold-DC	0.099	0.871	0.028	0.996	0.009	0.999	0.009	0.998
Ours (Naive Implementation)	0.288	0.773	0.117	0.871	0.052	0.969	**0.058**	0.965
Ours (ControlNet E2E FT Inpainting)	0.239	**0.808**	0.103	0.899	**0.049**	**0.972**	0.103	**0.972**

Chapter 6

Discussion

Our experiments demonstrate that augmenting diffusion-based monocular depth estimators with sparse depth conditioning through a ControlNet substantially improves affine-invariant accuracy while retaining deterministic single-step inference. The results highlight several important principles about how diffusion-based models respond to sparse depth guidance.

The improvements observed across indoor datasets indicate that latent diffusion models can effectively use sparse depth maps to refine their predictions even when trained without real depth data. The inpainting variant, which introduces latent-domain conditioning and explicit validity masks, shows that aligning the sparse-depth conditioning pathway with the prediction domain is essential for consistent integration of sparse measurements. The strong indoor performance suggests that the pretrained diffusion priors already encode reliable scene structure, and that sparse depth guidance acts as a local corrective signal rather than requiring large architectural changes.

The weaker performance on KITTI reveals a domain-dependence typical of diffusion-based estimators trained primarily on synthetic indoor data. While affine-invariant normalization mitigates scale differences, it cannot compensate for the texture bias and narrower depth distributions of indoor datasets. The discrepancy therefore reflects the limited exposure of the pretrained model to outdoor visual statistics rather than a limitation of the sparse-conditioning mechanism itself. Importantly, the single-step inference design allows clear attribution of these effects to data and conditioning rather than to stochastic sampling noise.

Relative to Marigold and E2E-FT, the presented framework extends the deterministic diffusion paradigm by introducing a dedicated sparse depth conditioning branch. Unlike Marigold-DC, which relies on iterative test-time optimization to align predicted and sparse depths, the proposed model learns this relationship directly through supervised training. This eliminates per-image optimization while achieving comparable structural fidelity, confirming that sparse depth guidance can be incorporated within a feed-forward diffusion model rather than recovered through gradient-based alignment at inference. The resulting runtime reduction is substantial: the proposed method completes inference in a single forward pass, while Marigold-DC typically requires dozens of optimization iterations.

Compared with discriminative baselines such as Metric3D and Depth Anything, the ControlNet models trade absolute metric precision for flexibility and data efficiency. Metric3D achieves lower AbsRel values on KITTI due to its supervised metric training, but its reliance on labeled data and camera calibration contrasts with the affine-invariant and self-supervised nature of this work. These results therefore bridge the gap between generative and discriminative depth estimation, showing that pretrained diffusion backbones can serve as competitive zero-shot depth predictors once augmented with sparse depth conditioning.

The findings suggest that conditioning latent diffusion models on sparse depth maps provides an efficient way to incorporate partial geometric information without requiring stochastic sampling or test-time refine-

ment. Because the conditioning operates in latent space and is limited to a single forward pass, the approach scales well to large datasets and real-time applications such as SLAM initialization, robotics perception, or AR reconstruction, where partial depth measurements are available. The same formulation could also be extended to handle other structured measurements such as LiDAR or depth-from-stereo data by adapting the control input format.

The results further imply that diffusion-based estimators benefit from combining pretrained image priors with sparse depth supervision: the diffusion backbone captures global structure, while the sparse depth signal enforces local metric consistency. This division of roles may guide future work toward hybrid systems that integrate pretrained diffusion backbones with physically grounded sparse measurements. Extending training to real or semi-synthetic outdoor data with pseudo-depth labels (as seen in Depth Anything v2 [99]) would likely close the remaining domain gap on KITTI-style benchmarks.

While the proposed approach achieves strong indoor performance and large speedups over optimization-based completion, its accuracy still lags behind fully supervised metric estimators. The reliance on affine-invariant normalization prevents recovery of absolute scale, and the current training datasets limit generalization to scenes with extreme depth ranges or lighting conditions. Additionally, single-step inference constrains the expressive capacity of the model; multi-step variants with learned weighting of sparse information could yield further improvements at moderate computational cost.

Overall, this work demonstrates that explicit sparse depth conditioning enables pretrained latent diffusion models to perform guided monocular depth estimation efficiently and consistently. By replacing test-time optimization with a learned conditioning pathway, the method advances the practicality of diffusion-based depth estimation and establishes a foundation for future work on fast, feed-forward depth completion with diffusion backbones.

Chapter 7

Conclusion

This thesis studied guided monocular depth estimation with sparse depth maps as conditions in a latent diffusion framework. Starting from the E2E-FT backbone, we attached a ControlNet to inject sparse measurements into the denoiser. We examined a naïve variant that conditions on the sparse depth image in pixel space and an inpainting variant that conditions in the VAE latent space using the RGB latent, an encoded sparse-depth latent, and a binary validity mask, trained with a composite loss that couples the affine-invariant objective with a sparse-consistency term. Throughout, the VAE and U-Net remained frozen and inference used a single deterministic step.

The main conclusions are as follows.

1) Explicit sparse conditioning improves diffusion-based MDE without sacrificing efficiency. Across NYUv2, ScanNet, and DIODE, the inpainting variant achieves lower AbsRel and higher δ_1 than diffusion MDE baselines under the same zero-shot evaluation protocol, while keeping single-pass inference. The gains over the naïve variant confirm that conditioning in the latent domain with an explicit validity mask is important for the ControlNet to influence features that determine the output.

2) Conditioning must be aligned with the prediction domain and made compulsory by the loss. The naïve variant delivered only marginal gains because the frozen backbone can minimize the affine-invariant loss through the $[\mathbf{x}, \mathbf{0}]$ path, leaving the ControlNet residuals near zero. Moving the condition to latent space, preserving resolution in the ControlNet embedding, and adding a sparse-consistency term created a direct supervision path at measured pixels and produced consistent improvements.

3) The method closes much of the gap to diffusion MDE, but dedicated DC remains more accurate. When compared to Marigold-DC on the same datasets, our model does not yet match its accuracy. Marigold-DC performs iterative test-time optimization and benefits from many denoising steps, while our model is fully feed-forward. The comparison highlights a clear accuracy–efficiency trade-off: our pipeline gives deterministic outputs in one step and therefore offers large speedups, which is attractive for real-time use, but top accuracy under sparse guidance still favors multi-step optimization.

4) Domain coverage and supervision drive the remaining error. Performance on KITTI lags indoor datasets, reflecting the outdoor domain gap of the training data and the limits of an affine-invariant objective. Discriminative metric estimators trained with camera-aware supervision retain an advantage on driving scenes. This points to data and supervision, rather than model capacity, as the main bottlenecks.

Implications. The results show that pretrained latent diffusion backbones can be augmented with sparse-depth conditioning to yield fast, deterministic guided predictions. The ControlNet inpainting formulation provides a practical alternative to test-time optimization for applications that require stable latency, such as robotics perception, AR scene understanding, or SLAM initialization where sparse but reliable depth is available.

Limitations. The current model is affine-invariant and does not recover absolute scale. It was trained primarily on synthetic data and shows a measurable domain gap outdoors. The single-step constraint simplifies training and inference but reduces the opportunity to iteratively absorb guidance in challenging regions. The sparse-consistency term improves local agreement but does not guarantee exact reproduction at measured pixels.

Future work. Several extensions follow directly from the analysis in Chapter 6. (i) Incorporate real outdoor data or high-quality pseudo-depth to reduce domain shift while keeping affine-invariant training. (ii) Add lightweight metric heads or camera-aware normalization to recover absolute scale at test time. (iii) Explore two to four denoising steps with learned step weights to trade a small cost for higher fidelity near sparse points. (iv) Replace the pixel-space sparse penalty with a latent-space counterpart or a mask-aware decoder loss to tighten supervision at measured pixels. (v) Unfreeze a small subset of U-Net layers or use LoRA adapters [33] to increase capacity for sparse-guided adjustments without losing the pretrained prior. (vi) Evaluate SfM point inputs and irregular LiDAR patterns to validate robustness to real sparsity structures and noise. (vii) Extend to video with temporal smoothing or recurrent conditioning to stabilize predictions over time.

In summary, attaching a ControlNet to an E2E-FT depth backbone and conditioning in latent space with an explicit validity mask yields a simple, fast, and effective guided MDE model. It improves over monocular diffusion baselines on standard zero-shot benchmarks while keeping one-pass inference. Closing the residual gap to dedicated depth completion methods appears feasible through modest increases in step count, richer supervision, and broader training data, without giving up the efficiency advantages established in this work.

Appendix A

Appendix

A.1 Full Experiment Results

A.1.1 Full Results on ScanNet

In this section, we present the full quantitative results on the ScanNet validation set, including all evaluation metrics used throughout this work. These results extend the main text by providing additional metrics such as squared relative difference, RMSE_{log}, $\log_{10}$, inverse RMSE, and scale-invariant RMSE.

Method	**AbsRel**	**SqRel**	**RMSE**	**RMSE_{log}**	**$\log_{10}$**	δ_1	δ_2	δ_3	**iRMSE**	**SIlog**
Marigold E2E FT	0.0581	0.0183	0.1545	0.0860	0.0244	0.9620	0.9891	0.9971	0.0690	8.57
Marigold	0.0642	0.0211	0.1666	0.0920	0.0271	0.9521	0.9880	0.9969	0.1012	9.16
Lotus-D	0.0549	0.0177	0.1672	0.0840	0.0240	0.9653	0.9904	0.9972	0.0576	8.38
Lotus-G	0.0592	0.0202	0.1763	0.0895	0.0259	0.9573	0.9884	0.9967	0.0621	8.92
SD E2E FT	0.0579	0.0175	0.1535	0.0858	0.0244	0.9648	0.9905	0.9974	0.0931	8.54
Depth Anything	0.0903	0.0322	0.2087	0.2093	0.0461	0.9161	0.9754	0.9877	14.27	20.80
Depth Anything v2	0.0924	0.0350	0.2162	0.2184	0.0474	0.9140	0.9726	0.9863	15.32	21.69
Metric3D v2-S	0.0323	–	0.1117	–	–	0.9887	0.9976	0.9992	–	0.0241
Metric3D v1 large	0.1032	–	0.2688	–	–	0.8761	0.9736	0.9953	–	0.0598
Ours (Final Model)	0.0475	0.0120	0.1353	0.0737	0.0204	0.9776	0.9943	0.9983	0.0547	7.35
Marigold-DC	0.0090	0.0041	0.1305	0.0223	0.0039	0.9977	0.9994	0.9998	0.0046	1.89

Table A.1: Full quantitative results on the **ScanNet** validation set. The table reports ten standard depth estimation metrics, including absolute and squared relative differences, various RMSE variants, $\log_{10}$ error, and accuracy under thresholds δ_1–δ_3.

A.1.2 Full Results on NYUv2

This section reports the full quantitative results on the NYUv2 validation set. As with the ScanNet results, we provide all ten metrics to enable detailed comparison across baseline and proposed models, including squared relative error, RMSE_{log}, $\log_{10}$ error, inverse RMSE, and scale-invariant RMSE.

A.1.3 Full Results on KITTI

This section presents the complete quantitative results on the KITTI Eigen test split. As with the other datasets, we include all ten evaluation metrics for a comprehensive comparison, covering absolute and

Method	AbsRel	SqRel	RMSE	RMSE_{log}	log_{10}	δ_1	δ_2	δ_3	iRMSE	SIlog
Marigold E2E FT	0.0524	0.0229	0.2178	0.0790	0.0224	0.9656	0.9916	0.9977	0.0427	7.87
Marigold	0.0554	0.0235	0.2245	0.0819	0.0238	0.9636	0.9913	0.9978	0.0444	8.15
Lotus-D	0.0512	0.0248	0.2496	0.0793	0.0226	0.9723	0.9913	0.9969	0.0364	7.90
Lotus-G	0.0538	0.0269	0.2581	0.0818	0.0236	0.9674	0.9916	0.9972	0.0379	8.15
SD E2E FT	0.0542	0.0234	0.2173	0.0808	0.0231	0.9649	0.9906	0.9976	0.0446	8.04
Depth Anything	0.1170	0.0703	0.3804	0.3363	0.0663	0.8726	0.9604	0.9782	30.52	33.31
Depth Anything v2	0.1207	0.0733	0.3857	0.3729	0.0706	0.8649	0.9588	0.9766	36.16	36.89
Metric3D v1 large	0.0663	–	0.2805	–	–	0.9477	0.9866	0.9960	–	0.0416
Ours (Final Model)	0.0492	0.0190	0.1991	0.0736	0.0211	0.9723	0.9932	0.9982	0.0403	7.33
Marigold-DC	0.0094	0.0031	0.1284	–	–	0.9987	0.9998	0.9999	–	1.89

Table A.2: Full quantitative results on the **NYUv2** validation set. Reported metrics include absolute and squared relative errors, RMSE variants, log_{10} error, and accuracy thresholds δ_1–δ_3.

squared relative errors, RMSE variants, log_{10} error, accuracy thresholds, and scale-invariant metrics.

Method	AbsRel	SqRel	RMSE	RMSE_{log}	log_{10}	δ_1	δ_2	δ_3	iRMSE	SIlog
Marigold E2E FT	0.0963	0.3992	3.1986	0.1405	0.0416	0.9187	0.9871	0.9964	88.05	13.97
Marigold	0.0985	0.4247	3.2986	0.1389	0.0421	0.9159	0.9872	0.9964	53.26	13.81
Lotus-D	0.0812	0.4222	3.7736	0.1205	0.0355	0.9310	0.9860	0.9970	0.0081	11.97
Lotus-G	0.0852	0.4622	4.0089	0.1276	0.0375	0.9221	0.9837	0.9964	0.0084	12.66
SD E2E FT	0.0960	0.3967	3.2111	0.1345	0.0413	0.9211	0.9883	0.9969	47.50	13.38
Depth Anything	0.3091	1.9514	6.7078	2.2616	0.3212	0.4542	0.8087	0.8892	16428.51	221.40
Depth Anything v2	0.3071	1.9223	6.6518	2.2689	0.3202	0.4547	0.8137	0.8906	16520.50	222.16
Metric3D v1 large	0.0580	–	2.7633	–	–	0.9639	0.9932	0.9982	–	0.0381
Ours (Final Model)	0.1028	0.4263	3.4394	0.1494	0.0453	0.8991	0.9834	0.9956	64.84	14.84
Marigold-DC	0.0282	0.0351	0.9580	–	–	0.9963	0.9995	0.9999	–	4.46

Table A.3: Full quantitative results on the **KITTI** Eigen test split. Results include ten standard metrics for depth estimation, enabling a detailed comparison across methods.

A.1.4 Full Results on DIODE

This section reports the full quantitative results on the DIODE validation set. As with other datasets, we include all ten metrics to provide a detailed comparison across all evaluated methods. The results include both relative and absolute error measures, as well as scale-invariant and threshold-based accuracy metrics.

A.2 Naive Experiment Results

In this section, we present results from the *naive baseline experiment*, designed to evaluate the model performance at different checkpoints and levels of sparse input guidance. The naive model uses a simplified ControlNet configuration without additional regularization or architectural refinements. We compare two training checkpoints (0th and 20000th) while varying the number of sparse input points (20k, 40k, 80k, 160k, 320k). The aim is to assess how training progression and input sparsity influence reconstruction quality under otherwise identical settings.

Method	AbsRel	SqRel	RMSE	RMSE_{log}	log_{10}	δ_1	δ_2	δ_3	iRMSE	SIlog
Marigold E2E FT	0.3022	3.9050	3.7293	0.2914	0.0804	0.7788	0.8947	0.9407	0.0972	28.57
Marigold	0.3020	3.8719	3.7275	0.2949	0.0814	0.7759	0.8910	0.9401	0.1006	28.88
Lotus-D	0.2281	5.4404	5.6885	0.2995	0.0904	0.7380	0.8620	0.9206	0.0787	28.31
Lotus-G	0.2296	4.5820	5.6273	0.3057	0.0926	0.7294	0.8602	0.9198	0.0820	28.91
SD E2E FT	0.3031	3.8811	3.7066	0.2926	0.0814	0.7765	0.8941	0.9394	0.0974	28.70
Depth Anything	0.3706	4.3294	5.2882	0.4077	0.1217	0.6680	0.8155	0.8847	0.1801	40.30
Depth Anything v2	0.3730	4.3435	5.2929	0.4117	0.1230	0.6647	0.8146	0.8839	0.1834	40.71
Metric3D v1 large	0.4455	–	5.9210	–	–	0.7498	0.8902	0.9412	–	0.1381
Ours (Final Model)	0.2394	2.7377	4.1770	0.2688	0.0718	0.8077	0.9140	0.9512	0.0746	26.35
Marigold-DC	0.0990	0.0819	1.5299	0.1518	0.0416	0.8708	0.9761	0.9956	0.0853	15.03

Table A.4: Full quantitative results on the **DIODE** validation set. The table reports all standard evaluation metrics for depth estimation.

A.2.1 DIODE (Scale 1.0)

Table A.5 summarizes the quantitative results for the DIODE validation set under the naive setup. Metrics are reported for both the initial (0th) and trained (20000th) checkpoints, across multiple sparse depth sampling configurations.

Checkpoints	AbsRel	SqRel	RMSE	RMSE_{log}	log_{10}	δ_1	δ_2	δ_3	iRMSE	SIlog
0, 20000	0.2888	3.7627	4.6237	0.3008	0.0833	0.7714	0.8904	0.9372	0.0839	29.50
0, 40000	0.2888	3.7627	4.6237	0.3008	0.0833	0.7714	0.8904	0.9372	0.0839	29.50
0, 80000	0.2888	3.7627	4.6237	0.3008	0.0833	0.7714	0.8904	0.9372	0.0839	29.50
0, 160000	0.2888	3.7627	4.6237	0.3008	0.0833	0.7714	0.8904	0.9372	0.0839	29.50
0, 320000	0.2888	3.7627	4.6237	0.3008	0.0833	0.7714	0.8904	0.9372	0.0839	29.50
20000, 20000	0.2884	3.7759	4.6157	0.2995	0.0828	0.7728	0.8915	0.9376	0.0828	29.36
20000, 40000	0.2884	3.7759	4.6157	0.2995	0.0828	0.7728	0.8915	0.9376	0.0828	29.36
20000, 80000	0.2884	3.7759	4.6157	0.2995	0.0828	0.7728	0.8915	0.9376	0.0828	29.36
20000, 160000	0.2884	3.7759	4.6157	0.2995	0.0828	0.7728	0.8915	0.9376	0.0828	29.36
20000, 320000	0.2884	3.7759	4.6156	0.2995	0.0828	0.7728	0.8915	0.9376	0.0828	29.36

Table A.5: Naive model results on the **DIODE** validation set for different sparse input configurations. All experiments were conducted at Scale 1.0.

A.2.2 KITTI (Scale 1.0)

Table A.6 reports the naive model's quantitative performance on the KITTI Eigen test split. As with DIODE, the model is evaluated at the 0th and 20000th training checkpoints while varying the number of sparse input points. This analysis helps isolate the effects of both training progression and input sparsity under an otherwise identical architecture.

A.2.3 NYUv2 (Scale 1.0)

Table A.7 shows the quantitative results of the naive model on the NYUv2 validation set. As before, we compare the 0th and 20000th checkpoints while varying the number of sparse input points. This setup

Checkpoints	AbsRel	SqRel	RMSE	RMSE_{log}	log_{10}	δ_1	δ_2	δ_3	iRMSE	SIlog
0, 20000	0.1266	0.5755	3.9144	0.2102	0.0584	0.8450	0.9721	0.9907	358.84	20.81
0, 40000	0.1266	0.5755	3.9144	0.2102	0.0584	0.8450	0.9721	0.9907	358.84	20.81
0, 80000	0.1266	0.5755	3.9144	0.2102	0.0584	0.8450	0.9721	0.9907	358.84	20.81
0, 160000	0.1266	0.5755	3.9144	0.2102	0.0584	0.8450	0.9721	0.9907	358.84	20.81
0, 320000	0.1266	0.5755	3.9144	0.2102	0.0584	0.8450	0.9721	0.9907	358.84	20.81
20000, 20000	0.1167	0.5255	3.7006	0.1824	0.0525	0.8705	0.9769	0.9925	222.90	18.08
20000, 40000	0.1167	0.5255	3.7005	0.1824	0.0525	0.8706	0.9769	0.9925	222.96	18.08
20000, 80000	0.1167	0.5255	3.7005	0.1824	0.0525	0.8706	0.9769	0.9925	222.92	18.08
20000, 160000	0.1167	0.5255	3.7005	0.1824	0.0525	0.8706	0.9769	0.9925	222.92	18.08
20000, 320000	0.1167	0.5255	3.7005	0.1824	0.0525	0.8706	0.9769	0.9925	222.92	18.08

Table A.6: Naive model results on the **KITTI** Eigen test split for different sparse input configurations. All experiments were conducted at Scale 1.0.

allows for an isolated evaluation of the effects of training progression and input sparsity on indoor scene reconstruction.

Checkpoints	AbsRel	SqRel	RMSE	RMSE_{log}	log_{10}	δ_1	δ_2	δ_3	iRMSE	SIlog
0, 20000	0.0515	0.0239	0.2164	0.0778	0.0219	0.9691	0.9923	0.9976	0.0412	7.75
0, 40000	0.0515	0.0239	0.2164	0.0778	0.0219	0.9691	0.9923	0.9976	0.0412	7.75
0, 80000	0.0515	0.0239	0.2164	0.0778	0.0219	0.9691	0.9923	0.9976	0.0412	7.75
0, 160000	0.0515	0.0239	0.2164	0.0778	0.0219	0.9691	0.9923	0.9976	0.0412	7.75
0, 200000	0.0515	0.0239	0.2164	0.0778	0.0219	0.9691	0.9923	0.9976	0.0412	7.75
20000, 20000	0.0516	0.0240	0.2157	0.0776	0.0219	0.9691	0.9923	0.9977	0.0412	7.73
20000, 40000	0.0516	0.0240	0.2157	0.0776	0.0219	0.9691	0.9923	0.9977	0.0412	7.73
20000, 80000	0.0516	0.0240	0.2157	0.0776	0.0219	0.9691	0.9923	0.9977	0.0412	7.73
20000, 160000	0.0516	0.0240	0.2157	0.0776	0.0219	0.9691	0.9923	0.9977	0.0412	7.73
20000, 200000	0.0516	0.0240	0.2157	0.0776	0.0219	0.9691	0.9923	0.9977	0.0412	7.73

Table A.7: Naive model results on the **NYUv2** validation set for different sparse input configurations. All experiments were conducted at Scale 1.0.

A.2.4 ScanNet (Scale 1.0)

Table A.8 provides the naive model results on the ScanNet validation set. The model is evaluated at both the 0th and 20000th checkpoints while varying the number of sparse depth samples. This experiment follows the same structure as previous datasets, isolating the influence of checkpoint progression and sparse input density on indoor scene predictions.

Across all datasets, the naive model exhibits only marginal improvements between the 0th and 20000th checkpoints, indicating that early training provides slight gains in overall accuracy but does not substantially alter depth reconstruction performance. Furthermore, varying the number of sparse input points from 20k to 320k (or 200k for NYUv2 and ScanNet) produces negligible differences across all reported metrics. This suggests that, under a naive configuration and with a zero-initialization of the depth tensor, the model's capacity to leverage sparse input information remains limited and largely invariant to input density. These

Checkpoints	**AbsRel**	**SqRel**	**RMSE**	**RMSE$_{\log}$**	**log$_{10}$**	δ_1	δ_2	δ_3	**iRMSE**	**SIlog**
0, 20000	0.0573	0.0172	0.1530	0.0852	0.0243	0.9656	0.9910	0.9974	0.0731	8.50
0, 40000	0.0573	0.0172	0.1530	0.0852	0.0243	0.9656	0.9910	0.9974	0.0731	8.50
0, 80000	0.0573	0.0172	0.1530	0.0852	0.0243	0.9656	0.9910	0.9974	0.0731	8.50
0, 160000	0.0573	0.0172	0.1530	0.0852	0.0243	0.9656	0.9910	0.9974	0.0731	8.50
0, 200000	0.0573	0.0172	0.1530	0.0852	0.0243	0.9656	0.9910	0.9974	0.0731	8.50
20000, 20000	0.0583	0.0175	0.1550	0.0862	0.0247	0.9653	0.9908	0.9974	0.0716	8.60
20000, 40000	0.0583	0.0175	0.1550	0.0862	0.0247	0.9653	0.9908	0.9974	0.0716	8.60
20000, 80000	0.0583	0.0175	0.1550	0.0862	0.0247	0.9653	0.9908	0.9974	0.0716	8.60
20000, 160000	0.0583	0.0175	0.1550	0.0862	0.0247	0.9653	0.9908	0.9974	0.0716	8.60
20000, 200000	0.0583	0.0175	0.1550	0.0862	0.0247	0.9653	0.9908	0.9974	0.0716	8.60

Table A.8: Naive model results on the **ScanNet** validation set for different sparse input configurations. All experiments were conducted at Scale 1.0.

findings highlight the necessity of architectural and training refinements - introduced in later experiments - to better integrate sparse depth guidance and improve generalization across diverse datasets.

Bibliography

[1] Yogesh Balaji, Seungjun Nah, Xun Huang, Arash Vahdat, Jiaming Song, Qinsheng Zhang, Karsten Kreis, Miika Aittala, Timo Aila, Samuli Laine, Bryan Catanzaro, Tero Karras, and Ming-Yu Liu. ediff-i: Text-to-image diffusion models with an ensemble of expert denoisers, 2023.

[2] Luca Bartolomei, Matteo Poggi, Andrea Conti, Fabio Tosi, and Stefano Mattoccia. Revisiting depth completion from a stereo matching perspective for cross-domain generalization, 2023.

[3] Rumeysa Bodur, Erhan Gundogdu, Binod Bhattarai, Tae-Kyun Kim, Michael Donoser, and Loris Bazzani. iedit: Localised text-guided image editing with weak supervision, 2023.

[4] Tim Brooks, Aleksander Holynski, and Alexei A. Efros. Instructpix2pix: Learning to follow image editing instructions, 2023.

[5] Yohann Cabon, Naila Murray, and Martin Humenberger. Virtual kitti 2, 2020.

[6] Mingdeng Cao, Xintao Wang, Zhongang Qi, Ying Shan, Xiaohu Qie, and Yinqiang Zheng. Masactrl: Tuning-free mutual self-attention control for consistent image synthesis and editing, 2023.

[7] Zhe Cao, Gines Hidalgo, Tomas Simon, Shih-En Wei, and Yaser Sheikh. Openpose: Realtime multi-person 2d pose estimation using part affinity fields, 2019.

[8] Yun Chen, Bin Yang, Ming Liang, and Raquel Urtasun. Learning joint 2d-3d representations for depth completion, 2020.

[9] Xinjing Cheng, Peng Wang, Chenye Guan, and Ruigang Yang. Cspn++: Learning context and resource aware convolutional spatial propagation networks for depth completion, 2019.

[10] Xinjing Cheng, Peng Wang, and Ruigang Yang. Depth estimation via affinity learned with convolutional spatial propagation network, 2018.

[11] Jooyoung Choi, Sungwon Kim, Yonghyun Jeong, Youngjune Gwon, and Sungroh Yoon. Ilvr: Conditioning method for denoising diffusion probabilistic models, 2021.

[12] Hyungjin Chung, Jeongsol Kim, Michael T. Mccann, Marc L. Klasky, and Jong Chul Ye. Diffusion posterior sampling for general noisy inverse problems, 2024.

[13] Hyungjin Chung, Byeongsu Sim, Dohoon Ryu, and Jong Chul Ye. Improving diffusion models for inverse problems using manifold constraints, 2024.

[14] Guillaume Couairon, Jakob Verbeek, Holger Schwenk, and Matthieu Cord. Diffedit: Diffusion-based semantic image editing with mask guidance, 2022.

[15] Angela Dai, Angel X. Chang, Manolis Savva, Maciej Halber, Thomas Funkhouser, and Matthias Nießner. Scannet: Richly-annotated 3d reconstructions of indoor scenes, 2017.

[16] Prafulla Dhariwal and Alex Nichol. Diffusion models beat gans on image synthesis, 2021.

[17] Ainaz Eftekhar, Alexander Sax, Roman Bachmann, Jitendra Malik, and Amir Zamir. Omnidata: A scalable pipeline for making multi-task mid-level vision datasets from 3d scans, 2021.

[18] Tsu-Jui Fu, Wenze Hu, Xianzhi Du, William Yang Wang, Yinfei Yang, and Zhe Gan. Guiding instruction-based image editing via multimodal large language models, 2024.

[19] Rinon Gal, Yuval Alaluf, Yuval Atzmon, Or Patashnik, Amit H. Bermano, Gal Chechik, and Daniel Cohen-Or. An image is worth one word: Personalizing text-to-image generation using textual inversion, 2022.

[20] Gonzalo Martin Garcia, Karim Abou Zeid, Christian Schmidt, Daan de Geus, Alexander Hermans, and Bastian Leibe. Fine-tuning image-conditional diffusion models is easier than you think, 2025.

[21] A Geiger, P Lenz, C Stiller, and R Urtasun. Vision meets robotics: The kitti dataset. *The International Journal of Robotics Research*, 32(11):1231–1237, 2013.

[22] GonzaloMG. Stable diffusion e2e-ft depth. `https://huggingface.co/GonzaloMG/stable-diffusion-e2e-ft-depth`, 2024. Commit from September 13, 2024. Accessed: October 2025.

[23] Jakub Gregorek and Lazaros Nalpantidis. Steeredmarigold: Steering diffusion towards depth completion of largely incomplete depth maps, 2025.

[24] Shuyang Gu, Dong Chen, Jianmin Bao, Fang Wen, Bo Zhang, Dongdong Chen, Lu Yuan, and Baining Guo. Vector quantized diffusion model for text-to-image synthesis, 2022.

[25] Yuchao Gu, Xintao Wang, Jay Zhangjie Wu, Yujun Shi, Yunpeng Chen, Zihan Fan, Wuyou Xiao, Rui Zhao, Shuning Chang, Weijia Wu, Yixiao Ge, Ying Shan, and Mike Zheng Shou. Mix-of-show: Decentralized low-rank adaptation for multi-concept customization of diffusion models, 2023.

[26] Ming Gui, Johannes Schusterbauer, Ulrich Prestel, Pingchuan Ma, Dmytro Kotovenko, Olga Grebenkova, Stefan Andreas Baumann, Vincent Tao Hu, and Björn Ommer. Depthfm: Fast monocular depth estimation with flow matching, 2024.

[27] Haoyu Guo, He Zhu, Sida Peng, Haotong Lin, Yunzhi Yan, Tao Xie, Wenguan Wang, Xiaowei Zhou, and Hujun Bao. Multi-view reconstruction via sfm-guided monocular depth estimation, 2025.

[28] Jing He, Haodong Li, Wei Yin, Yixun Liang, Leheng Li, Kaiqiang Zhou, Hongbo Zhang, Bingbing Liu, and Ying-Cong Chen. Lotus: Diffusion-based visual foundation model for high-quality dense prediction, 2025.

[29] Amir Hertz, Ron Mokady, Jay Tenenbaum, Kfir Aberman, Yael Pritch, and Daniel Cohen-Or. Prompt-to-prompt image editing with cross attention control, 2022.

[30] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models, 2020.

[31] Jonathan Ho, Chitwan Saharia, William Chan, David J. Fleet, Mohammad Norouzi, and Tim Salimans. Cascaded diffusion models for high fidelity image generation, 2021.

[32] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance, 2022.

[33] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models, 2021.

[34] Inbar Huberman-Spiegelglas, Vladimir Kulikov, and Tomer Michaeli. An edit friendly ddpm noise space: Inversion and manipulations, 2024.

[35] Hugging Face. Ddimscheduler — diffusers documentation. `https://huggingface.co/docs/diffusers/en/api/schedulers/ddim`, 2025. Accessed: 2025-10-08.

[36] Lee Hyoseok, Kyeong Seon Kim, Kwon Byung-Ki, and Tae-Hyun Oh. Zero-shot depth completion via test-time alignment with affine-invariant depth prior, 2025.

[37] Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. Elucidating the design space of diffusion-based generative models, 2022.

[38] Tero Karras, Samuli Laine, and Timo Aila. A style-based generator architecture for generative adversarial networks, 2019.

[39] Tero Karras, Samuli Laine, Miika Aittala, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. Analyzing and improving the image quality of stylegan, 2020.

[40] Bahjat Kawar, Michael Elad, Stefano Ermon, and Jiaming Song. Denoising diffusion restoration models, 2022.

[41] Bahjat Kawar, Gregory Vaksman, and Michael Elad. Snips: Solving noisy inverse problems stochastically, 2021.

[42] Bahjat Kawar, Shiran Zada, Oran Lang, Omer Tov, Huiwen Chang, Tali Dekel, Inbar Mosseri, and Michal Irani. Imagic: Text-based real image editing with diffusion models, 2023.

[43] Bingxin Ke, Anton Obukhov, Shengyu Huang, Nando Metzger, Rodrigo Caye Daudt, and Konrad Schindler. Repurposing diffusion-based image generators for monocular depth estimation, 2024.

[44] Bingxin Ke, Kevin Qu, Tianfu Wang, Nando Metzger, Shengyu Huang, Bo Li, Anton Obukhov, and Konrad Schindler. Marigold: Affordable adaptation of diffusion-based image generators for image analysis, 2025.

[45] Diederik P Kingma and Max Welling. Auto-encoding variational bayes, 2022.

[46] Peter Kocsis, Julien Philip, Kalyan Sunkavalli, Matthias Nießner, and Yannick Hold-Geoffroy. Lightit: Illumination modeling and control for diffusion models, 2024.

[47] Ang Li, Zejian Yuan, Yonggen Ling, Wanchao Chi, Shenghao Zhang, and Chong Zhang. A multi-scale guided cascade hourglass network for depth completion. In *2020 IEEE Winter Conference on Applications of Computer Vision (WACV)*, pages 32–40, 2020.

[48] Dongxu Li, Junnan Li, and Steven C. H. Hoi. Blip-diffusion: Pre-trained subject representation for controllable text-to-image generation and editing, 2023.

[49] Yuheng Li, Haotian Liu, Qingyang Wu, Fangzhou Mu, Jianwei Yang, Jianfeng Gao, Chunyuan Li, and Yong Jae Lee. Gligen: Open-set grounded text-to-image generation, 2023.

[50] Haotong Lin, Sida Peng, Jingxiao Chen, Songyou Peng, Jiaming Sun, Minghuan Liu, Hujun Bao, Jiashi Feng, Xiaowei Zhou, and Bingyi Kang. Prompting depth anything for 4k resolution accurate metric depth estimation, 2025.

[51] Yuankai Lin, Hua Yang, Tao Cheng, Wending Zhou, and Zhouping Yin. Dyspn: Learning dynamic affinity for image-guided depth completion. *IEEE Transactions on Circuits and Systems for Video Technology*, 34(6):4596–4609, 2024.

[52] Lina Liu, Xibin Song, Xiaoyang Lyu, Junwei Diao, Mengmeng Wang, Yong Liu, and Liangjun Zhang. Fcfr-net: Feature fusion based coarse-to-fine residual learning for depth completion, 2021.

[53] Nan Liu, Shuang Li, Yilun Du, Antonio Torralba, and Joshua B. Tenenbaum. Compositional visual generation with composable diffusion models, 2023.

[54] Zhiheng Liu, Ka Leong Cheng, Qiuyu Wang, Shuzhe Wang, Hao Ouyang, Bin Tan, Kai Zhu, Yujun Shen, Qifeng Chen, and Ping Luo. Depthlab: From partial to complete, 2024.

[55] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization, 2019.

[56] Andreas Lugmayr, Martin Danelljan, Andres Romero, Fisher Yu, Radu Timofte, and Luc Van Gool. Repaint: Inpainting using denoising diffusion probabilistic models, 2022.

[57] Ziwei Luo, Fredrik K. Gustafsson, Zheng Zhao, Jens Sjölund, and Thomas B. Schön. Image restoration with mean-reverting stochastic differential equations, 2023.

[58] Fangchang Ma, Guilherme Venturelli Cavalheiro, and Sertac Karaman. Self-supervised sparse-to-dense: Self-supervised depth completion from lidar and monocular camera, 2018.

[59] Fangchang Ma and Sertac Karaman. Sparse-to-dense: Depth prediction from sparse depth samples and a single image, 2018.

[60] Shweta Mahajan, Tanzila Rahman, Kwang Moo Yi, and Leonid Sigal. Prompting hard or hardly prompting: Prompt inversion for text-to-image diffusion models, 2023.

[61] Chenlin Meng, Yutong He, Yang Song, Jiaming Song, Jiajun Wu, Jun-Yan Zhu, and Stefano Ermon. Sdedit: Guided image synthesis and editing with stochastic differential equations, 2022.

[62] Ron Mokady, Amir Hertz, Kfir Aberman, Yael Pritch, and Daniel Cohen-Or. Null-text inversion for editing real images using guided diffusion models, 2022.

[63] Chong Mou, Xintao Wang, Liangbin Xie, Yanze Wu, Jian Zhang, Zhongang Qi, Ying Shan, and Xiaohu Qie. T2i-adapter: Learning adapters to dig out more controllable ability for text-to-image diffusion models, 2023.

[64] Alex Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models, 2021.

[65] Alex Nichol, Prafulla Dhariwal, Aditya Ramesh, Pranav Shyam, Pamela Mishkin, Bob McGrew, Ilya Sutskever, and Mark Chen. Glide: Towards photorealistic image generation and editing with text-guided diffusion models, 2022.

[66] Jinsun Park, Kyungdon Joo, Zhe Hu, Chi-Kuei Liu, and In So Kweon. Non-local spatial propagation network for depth completion, 2020.

[67] Gaurav Parmar, Krishna Kumar Singh, Richard Zhang, Yijun Li, Jingwan Lu, and Jun-Yan Zhu. Zero-shot image-to-image translation, 2023.

[68] Jiaxiong Qiu, Zhaopeng Cui, Yinda Zhang, Xingdi Zhang, Shuaicheng Liu, Bing Zeng, and Marc Pollefeys. Deeplidar: Deep surface normal guided depth prediction for outdoor scene from sparse lidar data and single color image, 2019.

[69] Aditya Ramesh, Prafulla Dhariwal, Alex Nichol, Casey Chu, and Mark Chen. Hierarchical text-conditional image generation with clip latents, 2022.

[70] René Ranftl, Alexey Bochkovskiy, and Vladlen Koltun. Vision transformers for dense prediction, 2021.

[71] René Ranftl, Katrin Lasinger, David Hafner, Konrad Schindler, and Vladlen Koltun. Towards robust monocular depth estimation: Mixing datasets for zero-shot cross-dataset transfer, 2020.

[72] Hareesh Ravi, Sachin Kelkar, Midhun Harikumar, and Ajinkya Kale. Preditor: Text guided image editing with diffusion prior, 2023.

[73] Kyeongha Rho, Jinsung Ha, and Youngjung Kim. Guideformer: Transformers for image guided depth completion. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6240–6249, 2022.

[74] Mike Roberts, Jason Ramapuram, Anurag Ranjan, Atulit Kumar, Miguel Angel Bautista, Nathan Paczan, Russ Webb, and Joshua M. Susskind. Hypersim: A photorealistic synthetic dataset for holistic indoor scene understanding, 2021.

[75] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models, 2022.

[76] Nataniel Ruiz, Yuanzhen Li, Varun Jampani, Yael Pritch, Michael Rubinstein, and Kfir Aberman. Dreambooth: Fine tuning text-to-image diffusion models for subject-driven generation, 2023.

[77] Chitwan Saharia, William Chan, Huiwen Chang, Chris A. Lee, Jonathan Ho, Tim Salimans, David J. Fleet, and Mohammad Norouzi. Palette: Image-to-image diffusion models, 2022.

[78] Chitwan Saharia, William Chan, Saurabh Saxena, Lala Li, Jay Whang, Emily Denton, Seyed Kamyar Seyed Ghasemipour, Burcu Karagol Ayan, S. Sara Mahdavi, Rapha Gontijo Lopes, Tim Salimans, Jonathan Ho, David J Fleet, and Mohammad Norouzi. Photorealistic text-to-image diffusion models with deep language understanding, 2022.

[79] Chitwan Saharia, Jonathan Ho, William Chan, Tim Salimans, David J. Fleet, and Mohammad Norouzi. Image super-resolution via iterative refinement, 2021.

[80] Tim Salimans and Jonathan Ho. Progressive distillation for fast sampling of diffusion models, 2022.

[81] Christoph Schuhmann, Romain Beaumont, Richard Vencu, Cade Gordon, Ross Wightman, Mehdi Cherti, Theo Coombes, Aarush Katta, Clayton Mullis, Mitchell Wortsman, Patrick Schramowski, Srivatsa Kundurthy, Katherine Crowson, Ludwig Schmidt, Robert Kaczmarczyk, and Jenia Jitsev. Laion-5b: An open large-scale dataset for training next generation image-text models, 2022.

[82] Nathan Silberman, Derek Hoiem, Pushmeet Kohli, and Rob Fergus. Indoor segmentation and support inference from rgbd images. In Andrew Fitzgibbon, Svetlana Lazebnik, Pietro Perona, Yoichi Sato, and Cordelia Schmid, editors, *Computer Vision – ECCV 2012*, pages 746–760, Berlin, Heidelberg, 2012. Springer Berlin Heidelberg.

[83] Jascha Sohl-Dickstein, Eric A. Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics, 2015.

[84] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models, 2022.

[85] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations, 2021.

[86] Xuan Su, Jiaming Song, Chenlin Meng, and Stefano Ermon. Dual diffusion implicit bridges for image-to-image translation, 2023.

[87] Jie Tang, Fei-Peng Tian, Wei Feng, Jian Li, and Ping Tan. Learning guided convolutional network for depth completion, 2019.

[88] Narek Tumanyan, Michal Geyer, Shai Bagon, and Tali Dekel. Plug-and-play diffusion features for text-driven image-to-image translation, 2022.

[89] Jonas Uhrig, Nick Schneider, Lukas Schneider, Uwe Franke, Thomas Brox, and Andreas Geiger. Sparsity invariant cnns, 2017.

[90] Aaron van den Oord, Nal Kalchbrenner, Oriol Vinyals, Lasse Espeholt, Alex Graves, and Koray Kavukcuoglu. Conditional image generation with pixelcnn decoders, 2016.

[91] Igor Vasiljevic, Nick Kolkin, Shanyi Zhang, Ruotian Luo, Haochen Wang, Falcon Z. Dai, Andrea F. Daniele, Mohammadreza Mostajabi, Steven Basart, Matthew R. Walter, and Gregory Shakhnarovich. Diode: A dense indoor and outdoor depth dataset, 2019.

[92] Massimiliano Viola, Kevin Qu, Nando Metzger, Bingxin Ke, Alexander Becker, Konrad Schindler, and Anton Obukhov. Marigold-dc: Zero-shot monocular depth completion with guided diffusion, 2025.

[93] Tengfei Wang, Ting Zhang, Bo Zhang, Hao Ouyang, Dong Chen, Qifeng Chen, and Fang Wen. Pretraining is all you need for image-to-image translation, 2022.

[94] Yufei Wang, Bo Li, Ge Zhang, Qi Liu, Tao Gao, and Yuchao Dai. Lrru: Long-short range recurrent updating networks for depth completion, 2023.

[95] Chen Henry Wu and Fernando De la Torre. A latent space of stochastic diffusion models for zero-shot image editing and guidance. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 7378–7387, October 2023.

[96] Qiucheng Wu, Yujian Liu, Handong Zhao, Ajinkya Kale, Trung Bui, Tong Yu, Zhe Lin, Yang Zhang, and Shiyu Chang. Uncovering the disentanglement capability in text-to-image diffusion models, 2022.

[97] Xin Xiong, Haipeng Xiong, Ke Xian, Chen Zhao, Zhiguo Cao, and Xin Li. Sparse-to-dense depth completion revisited: Sampling strategy and graph construction. In Andrea Vedaldi, Horst Bischof, Thomas Brox, and Jan-Michael Frahm, editors, *Computer Vision – ECCV 2020*, pages 682–699, Cham, 2020. Springer International Publishing.

[98] Lihe Yang, Bingyi Kang, Zilong Huang, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. Depth anything: Unleashing the power of large-scale unlabeled data, 2024.

[99] Lihe Yang, Bingyi Kang, Zilong Huang, Zhen Zhao, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. Depth anything v2, 2024.

[100] Hu Ye, Jun Zhang, Sibo Liu, Xiao Han, and Wei Yang. Ip-adapter: Text compatible image prompt adapter for text-to-image diffusion models, 2023.

[101] Wei Yin, Chi Zhang, Hao Chen, Zhipeng Cai, Gang Yu, Kaixuan Wang, Xiaozhi Chen, and Chunhua Shen. Metric3d: Towards zero-shot metric 3d prediction from a single image, 2023.

[102] Wei Yin, Jianming Zhang, Oliver Wang, Simon Niklaus, Long Mai, Simon Chen, and Chunhua Shen. Learning to recover 3d scene shape from a single image, 2020.

[103] Zhang Youmin, Guo Xianda, Poggi Matteo, Zhu Zheng, Huang Guan, and Mattoccia Stefano. Completionformer: Depth completion with convolutions and vision transformers, 2023.

[104] Zhu Yu, Zehua Sheng, Zili Zhou, Lun Luo, Si-Yuan Cao, Hong Gu, Huaqi Zhang, and Hui-Liang Shen. Aggregating feature point cloud for depth completion. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 8698–8709, 2023.

[105] Zheyuan Zhan, Defang Chen, Jian-Ping Mei, Zhenghe Zhao, Jiawei Chen, Chun Chen, Siwei Lyu, and Can Wang. Conditional image synthesis with diffusion models: A survey, 2025.

[106] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models, 2023.

[107] Shiwen Zhang, Shuai Xiao, and Weilin Huang. Forgedit: Text guided image editing via learning and forgetting, 2024.

[108] Bolei Zhou, Hang Zhao, Xavier Puig, Sanja Fidler, Adela Barriuso, and Antonio Torralba. Scene parsing through ade20k dataset. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5122–5130, 2017.

[109] Yiming Zuo and Jia Deng. Ogni-dc: Robust depth completion with optimization-guided neural iterations, 2024.